Who Bears the Burden? Commercial Property Taxation, Real Activity, and Revenue Implications

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Abstract

Property taxation is widely viewed as a promising revenue source for developing countries, yet evidence on its effects on real economic activity and distributional effects remains limited. This paper exploits a natural experiment in Kampala, Uganda, where a property revaluation and mass enumeration increased tax liabilities in the city while surrounding areas remained unaffected. Using administrative data from corporate and employee tax returns in a difference-in-differences framework, I document three main findings. First, the reform increased business rental expenses by 9 percent, with an estimated citywide pass-through rate of 0.65, contrary to the neoclassical prediction that landlords should bear nearly the entire burden. Second, the reform reduced business profits, investment, employment, and survival probability, while increasing wage informality. The effects are highly heterogeneous across firm size: revenue and profits decline for small firms, while large firms experience revenue gains, leaving their profits unchanged, suggesting they partially pass the burden to consumers through higher prices. Third, examining distributional consequences, I find that workers at the top of the earnings distribution were disproportionately affected, implying that the reform reduced within-firm wage inequality and was therefore progressive, a pattern consistent with skill downgrading within firms. Heterogeneity analysis reveals that financial constraints and leverage play a role in explaining the results.

JEL codes: H22, H24, H32, H71.
Key Words: Property Taxes, Firm Survival, Employment, Wages.

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I. INTRODUCTION

As fiscal space shrinks in many developing countries, governments are exploring ways to strengthen existing revenue instruments and develop new sources to generate additional revenue. Property taxation is widely viewed as holding significant untapped potential, with the opportunity to raise substantial revenue by increasing compliance from currently low levels.[1] In this context, recent literature examines the effectiveness of alternative policies and administrative strategies in improving property tax compliance and revenues (e.g., Manwaring and Regan 2025; Flores et al. 2025; Bergeron et al. 2024). However, evidence on the efficiency cost of property taxation and its distributional consequences remains limited.

While the revenue potential of property taxes can strengthen local governments' capacity to provide public goods and services that support a conducive business environment, such taxes also raise costs for local firms. For owner-occupiers, the burden is direct. Firms operating in rented premises may instead face higher rents if landlords pass the tax on to tenants. Consistent with this channel, recent studies document substantial pass-through of property taxes to rental costs (e.g., Baker 2024; Watson and Ziv 2024; Löffler and Siegloch 2021; Rolheiser 2019). These higher costs may affect firms in many ways. In this paper, I study the incidence of commercial property taxes at the firm level and examine their effects on firm survival, revenues, employment, wages, and the distributional consequences.

I exploit a natural experiment in Kampala, Uganda, where variations in property tax liabilities were introduced across different locations within the city through a revaluation and mass property enumeration exercise that began in 2016/17. Prior to this exercise, the last valuation had been conducted in 2006, meaning that taxable property values had remained unchanged for a decade. During this period, numerous new properties had been established that were not captured in the old tax roll. The 2016/17 exercise led to the creation of a new property tax roll, which included additional properties and assigned higher rateable values to existing ones. As a result, incumbent properties faced higher property tax liabilities due to increased valuations, while properties not previously on the tax roll experienced an extensive margin change, although the marginal tax rate remained unchanged in both cases. Following the exercise, the total number of properties on the tax roll increased by 176 percent, while potential revenue more than tripled.

This exercise was implemented in a staggered manner, beginning with one division

[1] In Uganda, it is estimated that only 10–11 percent of properties pay their annual property tax liability, a rate similar to that observed in many other cities in low-income countries (Manwaring and Regan 2025).

and gradually extending to others over time. For context, the capital city is divided into five administrative divisions. I exploit this geographical variation using a staggered Difference-in-Differences approach following Callaway and Sant'Anna (2021), with firms located outside the capital but within the Greater Kampala Metropolitan Area serving as never-treated units. This strategy allows me to credibly estimate the causal effects of increases in property tax liabilities on firm-level outcomes. The identifying assumption is that, in the absence of the reform, outcomes for firms in treated and untreated locations would have evolved similarly, i.e., the parallel trends assumption. I test this assumption using an event-study analysis. The analysis utilizes administrative tax data from corporate income and employee tax returns. These novel firm-level panel datasets span nine years (2013/14 - 2021/22) and were recently compiled by the Uganda Revenue Authority with support from UNU-WIDER.

As a "first stage" of the empirical analysis, I estimate the effect of the reform on business rental expenses and calculate the pass-through rate. I focus on firms in the balanced panel that reported paying rent in the year prior to the reform, estimating the effect of the reform on rent expenses scaled by their pre-reform values. I find that the reform led to a 9 percent increase in business rental expenses. This reduced form estimate implies that a UGX 1 increase in potential citywide property tax liabilities is associated with a UGX 0.42 increase in total rents paid by firms.[2] Interpreted as an incidence measure at the city level, this implies that approximately 42 percent of the aggregate commercial property tax burden is transmitted to firms through higher rents. Given that only about 65 percent of potential commercial property tax liabilities are ultimately collected, this estimate further implies that a UGX 1 increase in collected citywide property tax revenue is associated with a UGX 0.65 increase in total rents paid by firms. This result contrast the neoclassical view which asserts that the burden property taxes should fall almost entirely on the landlords. To explain this contrast, I propose a framework that embeds a key institutional feature of selective enforcement: the probability of enforcement, and therefore compliance, is higher when a property is occupied and generating observable economic activity, and lower when the unit is unoccupied. Under these conditions, property taxes raise the marginal cost of supplying space for rent.

I then analyze how the reform affected business revenues and profits, focusing on a panel of active firms. Overall, no significant effects are found on the revenues. However, the reform reduced firms' profits by 2.4 percentage points relative to pre-reform revenues,

[2] Because property taxes are levied at the property level and the data do not link tenant firms to landlords, I do not observe firm-specific property tax liabilities. As a result, it is not possible to construct a firm-level measure of tax pass-through.

while investment declined by 2.7 percentage points. The effects are very heterogeneous by firm size: revenue and profits decline for small firms, while large firms experience revenue gains, leaving their profits and investment unchanged. Although I can not cannot directly observe prices, this pattern is consistent with large firms exercising greater market power, enabling them to pass a share of the burden onto consumers through higher output prices. This size-dependent heterogeneity in pass-through capacity has important implications for the distributional incidence of commercial property taxation: the burden falls disproportionately on smaller, less financially resilient firms, while larger incumbents are relatively insulated.

To further understand firm dynamics, I examine the effects of the reform on firm survival. The survival outcome is constructed following the approach of Saez et al. (2019), restricting the sample to firms present in 2015/16 (the year prior to the reform) that reported non-zero turnover, regardless of whether they operated in other years. Firms are assigned a turnover value of zero in years in which they do not operate. The outcome variable is binary, equal to one if the firm operated in that year and zero otherwise. Consistent with Saez et al. (2019), the sample is reweighted using the nonparametric reweighting method developed by DiNardo et al. (1996) (DFL reweighting). The results indicate that the reform reduced firms' survival probability by 3.2 percent. Scaling this reduced-form effect by the aggregate increase in potential property tax liabilities implies that a $1 million increase in aggregate property tax liabilities leads to approximately 25 additional firm exits citywide.

A potential threat to my results is that they could be compounded by evasion of corporate income tax (CIT). To address this, I use a cross-tax validation strategy which classifies a case as exit if and only if the firm disappears from CIT and has no footprints in the employee tax returns. I further impose a stricter definition of exit, requiring no CIT return in year t and no record in employee tax returns in both years t and $t + 1$. Both of these restrictions show that the results are not driven by evasion responses. Moreover, heterogeneity analysis reveals that the negative survival effects are strongest among firms with no corporate income tax liability (i.e., loss-making firms). As an additional robustness check, the results are similar when the sample is restricted to firms that were present and reported positive turnover in the first year of the sample, rather than to firms in the year immediately prior to the reform.

Using the same sample, I analyze employment effects, coding employment as zero following firm exit and normalizing it relative to its pre-reform level to facilitate interpretation in percent changes. On average, the reform reduced employment by 4.6 percent among firms in treated areas relative to untreated areas. Scaling this reduced-form esti-

mate by the aggregate increase in potential property tax liabilities implies that a $1 million rise in aggregate liabilities is associated with approximately 170 fewer jobs citywide.[3] In a sample of firms active throughout the entire period, employment declines by 4 percent, indicating that the overall reduction in employment is not driven solely by firm exit but also by within-firm adjustments among continuing firms. As a robustness check, I re-estimate the employment effects using log and inverse hyperbolic sine transformation as well as levels.[4] Both the event-study plots and average effects are consistent with those obtained using normalized outcomes.

Consistent with the employment results, I find that the firms' wage bill declines by 6.3 percent following the reform. The results further reveal the reform led to an increase in informal wages by 3.3 percentage points from a pre-reform mean of 10.4 percent, suggesting that part of the observed decline in formal employment reflects jobs transitioning to informal arrangements rather than outright job destruction. This partly accounts for the large estimated employment effects.

Heterogeneity analysis reveals that survival and employment effects are concentrated among tenant firms and statistically insignificant for owner-occupiers. This pattern is consistent with substantial tax pass-through into business rental expenses, and with the fact that owner-occupiers tend to be larger and better positioned to absorb higher property tax liabilities by adjusting other margins. I also find that pre-reform profitability and leverage are important moderators. Effects are stronger among less profitable firms and monotonically decreasing in profitability. For leverage, I find a U-shaped pattern in tax incidence across the leverage distribution: high-leverage firms are vulnerable through a financial distress channel, whereby higher property tax obligations tighten liquidity against existing debt service; very low-leverage firms are vulnerable through a financial exclusion channel, whereby limited access to credit markets constrains their ability to smooth the shock, forcing adjustment on real margins. Firms with moderate leverage retain sufficient unused debt capacity to absorb the shock, explaining their relative insulation. I provide direct evidence corroborating this mechanism: moderately leveraged firms experience a differential increase in debt following the reform, while firms at either tail of the leverage distribution do not. This asymmetry suggests that access to the borrowing margin is the key mechanism insulating moderate-leverage firms from the employment and survival consequences of the reform.

Finally, I examine the distributional consequences of the reform at firm level by as-

[3] As with pass-through to rents, data limitations preclude calculation of firm-level employment elasticities with respect to property tax liabilities.

[4] The log transformation is log(1+employment) since employment can be zero in some years.

sessing which types of workers are most affected. The sample is restricted to active firms with at least five workers in the pre-reform period to ensure adequate support across the wage distribution. First, I examine how the share of workers, relative to pre-reform total employment, in fixed wage quartiles evolves following the reform. Second, I analyze how within-firm wage quantiles evolve before and after the reform in treated firms relative to untreated firms. The evidence indicates that workers at the top of the earnings distribution are disproportionately affected. In particular, following the reform, the share of workers in the top quartile declined by 2.8 percentage points relative to pre-reform total employment, while the share of workers in the 50th–75th percentile increased by 2.2 percentage points. The shares in other bins are unaffected. This pattern is consistent with skill downgrading within firms. Consistent with the above findings, I find that wages fall significantly at the top of the distribution while the no significant effects are found at the bottom of the distribution. These results suggest that the reform reduced within-firm wage inequality and was therefore progressive at firm level.

Event-study analyses show that the results on business rental expenses, firm survival, employment, and wages are not driven by pre-trends. The event-study plots indicate that pre-period coefficients are statistically indistinguishable from zero, suggesting the absence of differential pre-trends and that outcomes would have evolved similarly in treated and untreated areas in the absence of the reform.

This paper contributes to two distinct strands of the literature. First, it relates to research on how local taxes affect business decisions. Recent studies show that state and local tax policies in the United States influence business location choices (Fajgelbaum et al. 2019; Giroud and Rauh 2019; Rohlin et al. 2014) and employment decisions (Thompson and Rohlin 2012; Giroud and Rauh 2019). Far less attention, however, has been devoted to understanding how local taxes, particularly property taxes affect real activity and their distribution consequences. Moreover, the existing literature largely examines local sales taxes in developed-country contexts. By contrast, this paper focuses on property taxes in a developing-country setting. The effects of property taxes are likely to differ from those of sales taxes because the incidence falls on a different base. For example, the extent to which property taxes affect firms operating in rented premises depends on the pass-through rate from landlords to tenants. Most closely related to this paper is the recent work by Enami et al. (2023), which examines how school property taxes affect both the stock and flow of establishments in a location. However, their analysis focuses on a single type of property, schools, finding null effects.

I contribute to the literature by showing that commercial property taxes broadly affects business survival, employment, and wages. While no significant effects are found

among owner-occupier businesses, consistent with Enami et al. (2023) on school property taxes, substantial effects are uncovered among tenant businesses, driven by high tax pass-through. This implies that analyses focusing solely on property owners may substantially understate the true impact of property taxation on real activity. This paper also highlights financial constraints and financial distress as key mechanisms underlying these effects.

Second, this paper contributes to the long-standing literature on property tax incidence (see England (2016) and Oates and Fischel (2016) for reviews). While earlier work found low pass-through of property taxes to rent (Carroll and Yinger 1994), more recent studies have documented substantial pass-through to rental costs (Watson and Ziv 2024; Baker 2024), particularly when the supply of housing is elastic (Löffler and Siegloch 2021). Prior research, however, has focused primarily on rental housing. This paper provides evidence on pass-through in the context of commercial properties.

The remainder of the paper is organized as follows. Section II describes the institutional context, including property taxation arrangements in Uganda and details of the revaluation and property enumeration reform. Section III describes the corporate income tax and employee tax return data used in the analysis and presents descriptive statistics. Section IV outlines the identification strategy. Section V presents and discusses empirical evidence on the effects of the reform on business rental expenses and the associated theoretical contrasts. Section VI presents the empirical results on the effects of the reform on various dimensions of business activity, including revenues, firm survival, and employment, as well as the heterogeneity analysis. Section **??** examines the effects on wages and the associated distributional implications. Section IX concludes.

II. CONTEXT

II.A. Property Taxation in Uganda

Property taxes in Uganda are governed by the Local Government (Rating) Act, 2005, which allows local jurisdictions to levy rates, based on the rateable value of any property within their jurisdiction. The tax is levied on all immovable property, including buildings or structures of any kind that are commercially managed (e.g., rented residential houses, rented shops, factories, hotels). It also applies to owner-occupied properties used for business purposes but excludes vacant sites and owner-occupied residential properties.[5]

[5]The full list of exempted properties include: official residence of the president; official residences of cultural/traditional leaders; public worship places; cemeteries and crematoriums; charitable and educational institutions; outdoor sports and recreational facilities; local council-owned properties; property owned by privileged organizations under Diplomatic Privileges Act; International Treaty Tax-exempt organizations; and property belonging to any institution with which Government has contractual obligation not to levy

Local governments have the discretion to levy rates ranging from a minimum of 2,000 Uganda Shillings (approximately $0.60) up to 12 percent of the rateable value, where rateable value is defined as the assessed property value less a conservancy allowance of 22 percent. In Kampala Capital City Authority (KCCA), the property tax rate is set at 6 percent of the rateable value.

Currently, most cities (including the capital) and municipalities in Uganda use a market-value approach in which property rates are applied to the estimated annual rental value or business income value (Bakibinga and Ngabirano 2019). The law requires each local government to produce an initial valuation list and update it at least once every five years. In practice, however, these lists are often not updated on time, largely due to a shortage of qualified valuation surveyors in both the private and public sectors (Bakibinga and Ngabirano 2019) and inadequate resources.

II.B. Property Revaluation, Enumeration, and Tax Collection in Kampala

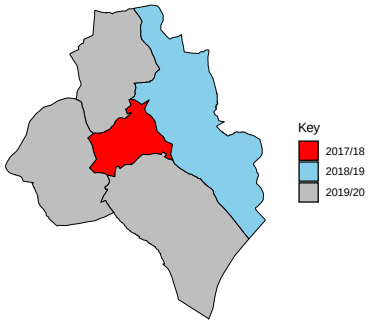
In an effort to increase revenue, the Kampala Capital City Authority (KCCA) initiated a revaluation and enumeration of properties beginning in the financial year 2016/17. Prior to this, the last valuation had been conducted in 2006, implying that the taxable values of properties had remained unchanged for a decade. Over the same period, numerous new properties had been developed that were not included in the old tax roll. The number of properties on the tax roll increased by 176 percent, from 127,021 to 223,329, while the average property tax liability increased by 90 percent after the exercise. The revaluation and enumeration exercise was therefore expected to substantially increase revenue for KCCA, where property tax had accounted for 35 percent of own-source revenue prior to the exercise. During the mass enumeration and surveying, KCCA collected detailed information on ownership, location (including GIS coordinates), and property attributes for valuation purposes. Valuation was conducted primarily by professional valuers, although in some cases it relied on regression model predictions.

This exercise was conducted in a staggered manner, starting with one division and gradually extending to others over time. Figure 1 shows the rollout of the new tax roll across the five divisions of the capital city, following the revaluation and enumeration exercise. The different colors in the figure represent the various phases of the exercise, as indicated in the legend.

The exercise resulted in the introduction of a new property tax roll that included more properties and assigned higher rateable values to existing ones. As a result, incumbent

fees and tax.

Panel A: Kampala Capital City



Panel B: Greater Kampala Metropolitan Area

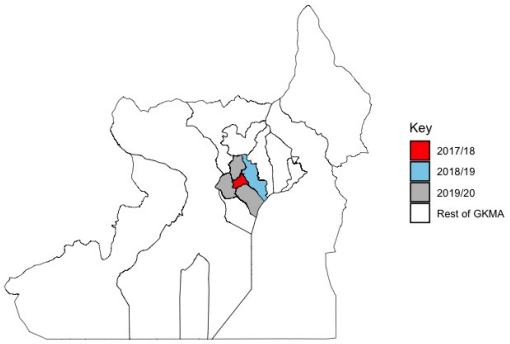


Figure 1: Kampala Capital City and Surrounding Areas

Notes: Panel A shows the five divisions of Kampala Capital City and the different phases of the rollout of the new property tax roll, as indicated in the legend. Panel B shows the Greater Kampala Metropolitan Area (GKMA), which includes counties in the neighborhood of the capital city. Firms in these locations serve as the never-treated units in the identification strategy, as these areas did not experience similar changes in property tax enforcement during the study period.

properties faced higher property tax liabilities due to increased valuations, while properties not previously on the tax roll experienced an extensive margin change, i.e., from zero to a positive tax liability, although the marginal tax rate remained unchanged in both cases. Figure 2 presents the number of properties and the revenue from each of the five divisions under both the old and new tax rolls. All divisions experienced an increase in the number of properties, with the number more than doubling in some divisions (Figure 2, Panel A). Similarly, potential revenue per property— a proxy for tax liability, increased across all divisions, with some experiencing more than 180% increase (Figure 2, Panel B).

Property Tax Collection:— After the valuation, initial valuation lists are displayed in hard copy at the respective division offices, with additional publication through the press, radio announcements, and local leaders. Property owners are then given one month to file objections, after which the valuation court convenes to hear all appeals. Decisions on the appeals may result in either an increase or a decrease in tax liability. Final valuation lists are then generated, and bills and demand notes are printed and delivered to property owners. Although the Authority is not able to physically deliver demand notes to every property owner, the digital system, eCitie, sends SMS and email notifications to all owners of the billed property.[6] KCCA is also legally authorized to issue a demand notice to a tenant if the property owner is unavailable, and the tenant is legally entitled to deduct

[6]Most of the information on the valuation and billing process was obtained from (Manwaring and Regan 2019).

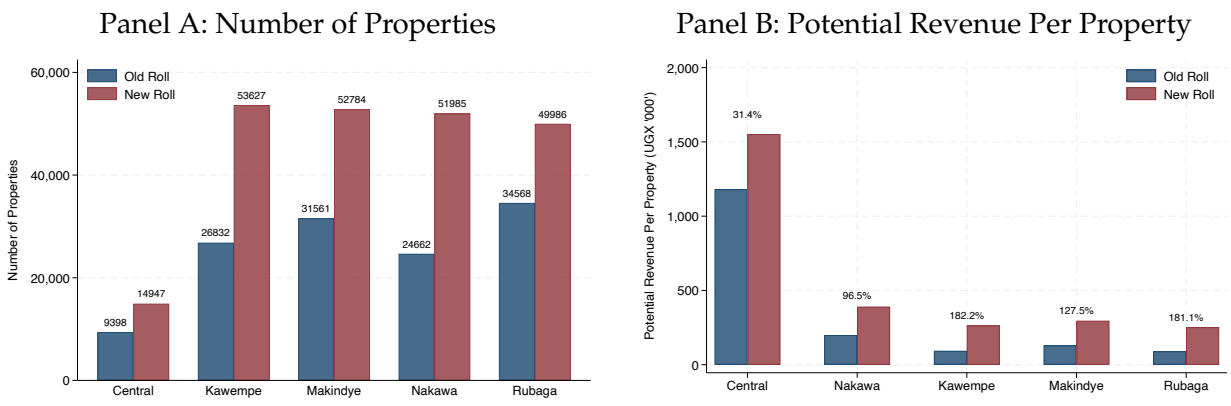


Figure 2: Number of Properties and Potential Property Tax Revenue

Notes: Panel A shows that number of properties in the old and new roll across the five divisions. Panel B shows the potential revenue per property before and after the reform. The data is extracted from Ahabwe et al. (2020) and Bakibinga and Ngabirano (2019).

the tax payment from their rent.

Billing for all properties included on the final valuation lists occurs at the beginning of the financial year (July 1). Taxpayers have until December 31 to pay their property taxes on time. They may request payment plans, which are officially limited to a maximum of four equal installments. If payment is not made in time, KCCA may issue a demand notice requiring the taxpayer to settle the outstanding rate, along with any accrued interest, within two months of the notice being served. Interest begins accruing 30 days after the end of the financial year, at a rate of 2 percent per month, for as long as the liability remains unpaid. Payments can be remitted through the bank or via mobile money.

III. DATA AND DESCRIPTIVE STATISTICS

III.A. Data Sources and Sample Construction

The paper utilizes administrative tax data from corporate income and employee tax returns as its primary datasets. The corporate income tax return (CIT) data comprise a firm-level panel covering the universe of firms registered for corporate income tax in Uganda, spanning the period 2013/14 – 2021/22. The employee tax return data cover the universe of individuals in formal employment over the same period. These are newly compiled dataset launched in 2022 by the Uganda Revenue Authority and UNU-WIDER.

The corporate income tax data contain annual balance sheets and profit and loss accounts, which allow me to observe detailed measures of business activity, including firms'

revenues, profits, and assets. All monetary values are recorded in Uganda shillings. To ensure comparability over time, I deflate them to constant 2016/17 values using the consumer price index (CPI). The data also include information on firms' sectors of operation and location. Location information is particularly important for defining treatment, as discussed later in the paper. Because the corporate income tax data do not contain information on employment, I link them to the employee tax return data to construct a measure of formal employment by counting the number of workers filing under each firm. Since employee tax returns are filed monthly while corporate income tax returns are filed annually, I compute average employment over the twelve months of each year.

Sample Construction:— The full dataset from corporate income tax returns comprises 404,527 firm–year observations. However, some firms report zero turnover throughout the entire sample period. After dropping these firms, the dataset is reduced to 299,683 firm–year observations, of which 81,162 (27 percent) form a balanced panel comprising 9,018 unique firms. The sample is further restricted to firms located in the Greater Kampala Metropolitan Area (GKMA) to ensure that never-treated firms are geographically as close as possible to treated firms. Because the reform affects only firms in the capital city, firms in surrounding locations, the rest of the GKMA (Figure 1, Panel B), serve as a plausible comparison group. This restriction yields 213,185 firm–year observations, representing 71 percent of firms that ever reported positive revenues during the sample period. Of these, 60,984 observations form a balanced panel comprising 6,776 unique firms, equivalent to 75 percent of the balanced-panel observations in the full dataset.

Additional sample restrictions vary by section or analysis. For the analysis of the effect on business rental expenses, the sample is restricted to firms that reported paying rent in the pre-reform period, as this outcome is normalized by its pre-reform value. For the baseline analyses of firm survival and employment effects, the sample is restricted to firms that operated and reported positive revenues in the pre-reform year.

III.B. Descriptive Statistics

Figure 3 plots the industry composition and size distribution across treated and comparison locations for firms that were active in the pre-reform year. The figure reveals substantial overlap along these dimensions. The size distribution is left-skewed: the majority of firms have annual revenues below UGX 150 million and fewer than five formal employees. However, the distribution is not systematically different for firms inside versus outside the city. Appendix Figure A.1 shows a similar overlap using density plots for revenues and assets. There are some notable differences in industry composition across sectors. For example, in retail and wholesale trade, the share of firms inside the city

exceeds that outside the city by a considerable margin, while the reverse holds for the education sector. I flexibly account for differences in size and industry composition using fixed effects in the analysis. There is sufficient overlap in both dimensions to allow for comparisons of firms within the same industry and size bins across treated and comparison locations.

Table 1 reports pre-reform (2015/16) means and standard deviations for selected variables across different groups. The average firm in the universe of corporate income tax (CIT) filers has annual revenues of UGX 556 million and total assets of UGX 267 million, of which 70 percent are fixed assets. Average revenues for treated and comparison firms exceed those in the CIT universe, but are broadly comparable across the two groups. In contrast, firms in comparison group hold more assets on average than treated firms, suggesting that firms in the city (treated locations) are more efficient in utilizing assets to generate revenue. Similarly, average formal employment is higher among comparison firms than among treated firms. To account for these size-related differences, I include in the regressions time-invariant size bins defined by pre-reform employment. Nonetheless, robustness checks confirm that results are insensitive to the inclusion or exclusion of size fixed effects.

Regarding renting or owning land and buildings, the average firm in the CIT universe pays annual rent of UGX 6.2 million and owns land and buildings worth UGX 72 million. Comparison firms have lower average rent expenses and are less likely to be renters (report positive rent) compared to treated firms. In contrast, comparison firms have higher average values of land and buildings and are more likely to own property (report positive values for land and buildings). To understand the implications of these differences, I disaggregated all my main results by whether firms are tenants or owner-occupiers.

Lastly, I present statistics on variables related to the financial situation of firms. Firms with better financial situations may respond differently to cost increases associated with property tax liabilities. Table 1 shows that the share of loss-making firms is balanced at 23 percent across comparison and treated firms and slightly lower in the CIT universe. Similarly, average profit margins (EBIT/revenue) are similar across comparison and treated firms. Treated firms have higher liquidity (cash-to-assets) relative to comparison firms. While leverage is slightly higher in the CIT universe at 41 percent, it is balanced across treated and comparison firms. Broadly, comparison and treated firms exhibit similar pre-reform financial situations along the dimensions of profitability and leverage.

Figure 3: Firm Size Distributions and Industry Composition

Notes: This figure plots the industry and size mix of pre-reform active firms located inside (treated) and outside (comparison) the city. Each graph's bars sum to 1 within each location. Panels A and B show the distribution of revenues and employment, respectively. Panel C shows the industry composition of firms in the sample following the International Standard Industrial Classification of All Economic Activities (ISIC, Rev.4). Utilities combines electricity, gas, steam, and air conditioning supply with water supply, sewerage, waste management, and remediation activities. Administration combines administrative and support service activities with public administration & defence and compulsory social security. I also exclude ISIC categories for activities of households as employers; undifferentiated goods- and services-producing activities of households for own use; and activities of extraterritorial organizations and bodies, as these account for negligible shares in the sample.

Table 1: Summary Statistics

	(1) Universe	(2) Ever-Treated	(3) Comparison Group
Revenues	556.09	614.96	629.81
	(1,232.30)	(1,294.14)	(1,328.00)
Total Assets	267.43	284.87	361.35
	(595.70)	(620.41)	(696.65)
Fixed Assets	188.27	190.70	294.15
	(461.14)	(468.84)	(568.66)
Formal Employees	5.16	5.15	7.45
	(10.69)	(10.43)	(13.00)
Rent Expenses	6.15	8.39	4.43
	(12.84)	(14.83)	(11.27)
Landing and Buildings	72.33	63.74	130.18
	(209.02)	(202.33)	(270.43)
$\mathbb{I}\{\text{Rent Expense} > 0\}$	0.44	0.50	0.34
	(0.50)	(0.50)	(0.47)
$\mathbb{I}\{\text{Landing and Buildings} > 0\}$	0.20	0.15	0.31
	(0.40)	(0.36)	(0.46)
$\mathbb{I}\{\text{Loss Making}\}$	0.20	0.23	0.23
	(0.40)	(0.42)	(0.42)
Profit Margins	0.15	0.13	0.14
	(0.31)	(0.31)	(0.33)
Cash-to-Assets	0.17	0.17	0.13
	(0.22)	(0.23)	(0.20)
Debt-to-Assets	0.41	0.35	0.36
	(0.36)	(0.35)	(0.35)

Note: Reports pre-reform (2015/16) means and standard deviations for selected variables across different groups. Standard deviations are reported in parentheses. Column 1 presents statistics for the CIT universe, while the remaining columns present statistics for different groups according to their treatment status. The ever-treated group consists of firms located in Kampala City, whereas the comparison group consists of firms from the rest of the GKMA. All monetary values are denominated in Uganda Shillings (UGX), constant 2016/17.

IV. IDENTIFICATION STRATEGY

I use difference-in-differences design to estimate the causal impact of property tax enforcement on business rental expenses, business activity, and wages. The identifying vari-

ation comes from comparing outcomes of firms located in Kampala versus those located in surrounding areas that did not experience a change in property tax enforcement and liability during the sample period. The identifying assumption is that, absent the reform, average outcomes for firms in treated and untreated areas would have evolved similarly (i.e., parallel trends assumption). I test this assumption using event-study analysis.

Traditionally, one would estimate the average and dynamic effects of the reform using Two-Way Fixed Effects (TWFE) difference-in-differences regressions, as in equations 1 and 2, respectively.

$$y_{it} = \alpha_g + \alpha_t + \beta D_{it} + \gamma X_{it} + \epsilon_{it}, \tag{1}$$

$$y_{it} = \alpha_g + \alpha_t + \sum_{\substack{e=-5 \\ e\neq -1}}^{4} \beta_e D_{it}^e + \gamma X_{it} + \nu_{it}, \tag{2}$$

where y_{it} is the outcome variable, i.e., business rental expenses, business activity measures, and wages. Detailed description of outcome measurement is presented in the respective sections, following this section. D_{it}^e is a time-varying indicator variable that is equal to one starting in the year firm i's location received the reform and zero otherwise. α_g and α_t are group fixed effects and year fixed effects. X_{it} is a vector of controls.

However, recent literature shows that under staggered adoption, TWFE regressions deliver consistent estimates only under homogeneous treatment effects. In the presence of heterogeneous effects across cohorts or over time, TWFE coefficients are generally biased and difficult to interpret (De Chaisemartin and d'Haultfoeuille 2020; Sun and Abraham 2021; Goodman-Bacon 2021; Callaway and Sant'Anna 2021; Borusyak et al. 2024). Bias in TWFE estimators arises from negative weighting induced by forbidden comparisons, whereby already-treated units are used as controls for later-treated units (Borusyak and Jaravel 2018). Heterogeneous effects are plausible in this setting for two reasons. First, firms' responses to changes in property tax enforcement and liability may evolve over time as they reoptimize along different margins. Second, changes in property tax liability vary across cohorts (Figure 2), implying treatment effects may differ by cohorts.

To circumvent the challenges associated with TWFE, I estimate treatment effects using the doubly robust estimator of Callaway and Sant'Anna (2021), using the not-yet-treated units as the comparison group. All the results are robust to using the never-treated firms as the comparison group. I also show that the main difference-in-differences and event-study estimates are robust when estimated using the estimators introduced in Sun and Abraham (2021) and Borusyak et al. (2024). Callaway and Sant'Anna (2021) show that this estimator is consistent in the presence of cohort-specific treatment effects. In addition, their approach allows for estimation and inference under conditional parallel trends, i.e.,

parallel trends assumption holds only after conditioning on observed covariates.

Rather than estimating dynamic effects via the event-study regression in equation 2, Callaway and Sant'Anna (2021) propose an aggregation scheme to highlights treatment effect heterogeneity by exposure length while avoiding the drawbacks associated with TWFE, as in equation 3.

$$\beta_{es}(e) = \sum_{g \in \mathcal{G}} \mathbb{I}\{g + e \leq \mathcal{T}\}\, \mathbb{P}(G = g \mid G + e \leq \mathcal{T})\, ATT(g, g + e), \tag{3}$$

where, e denotes event time (the number of periods since initial treatment), G denotes the first period a unit is treated, and $ATT(g,t)$ is the average treatment effect for units who are members of a particular group g at a particular time period t, i.e., $ATT(g,t) = \mathbb{E}[Y_t(g) - Y_t(0) \mid X, G_g = 1]$. $\beta_{es}(e)$ is therefore the average effect of participating in the treatment e periods after the treatment was adopted, averaging across all cohorts observed at event time e.

In estimation, I recognize that the the roll out of the reform was not randomly assigned across groups. The City Authority first targeted divisions with high-value properties, namely Central Division, followed by Nakawa (Figure 2, Panel B). Moreover, the never-treated firms are located outside the capital city and may differ systematically from ever-treated firms. As shown in Table 1, firms in the capital differ on observables from those outside the city. Accordingly, I include baseline controls, size bins, sector, and accounting month, in the regression specifications. I also present event-study evidence indicating no differential pre-trends between treated and comparison groups.

V. BUSINESS RENTAL EXPENSES AND OCCUPANCY STATUS

V.A. Effects on Business Rental Expenses

Given that a substantial share of businesses in the sample (about 45%) report paying rent in the pre-reform year, it is important to examine whether property taxes affect business rental costs, as a "first stage" of the empirical analysis. According to the neoclassical view, the burden of property taxation should fall on the least elastic factor, implying that landlords ultimately bear the tax incidence. If this prediction holds, property taxes should have no direct effect on business rental expenses. There is, however, limited empirical evidence testing this prediction in the context of commercial properties. Recent work focusing on rental housing provides evidence that contrasts with this prediction (e.g., Baker 2024).

I test whether the neoclassical prediction holds in the context of commercial proper-

ties. I consider firms in the balanced panel that reported paying rent in the year prior to the reform and estimate the effect of the reform on rent expenses, scaled by their pre-reform values. The results are presented in Figure 4. Prior to the reform, rent expenses for firms in locations that experienced an increase in property tax liability and those in unaffected areas followed similar trends, supporting the identifying assumption of parallel trends. Following the reform, firms in treated locations experienced a differential increase in rent expenses relative to those in untreated areas. The average effects presented Table 2 shows that the reform led to a 9 percent increase in business rental expenses (column 4).

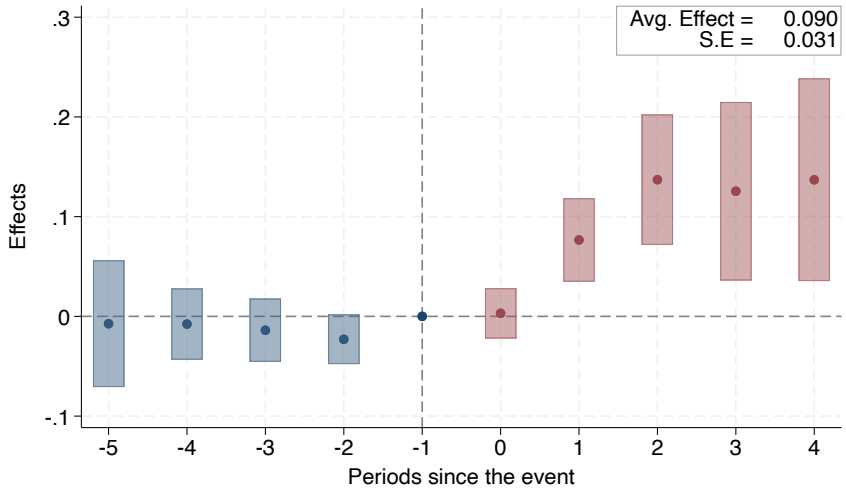


Figure 4: Effects of Property Taxes on Business Rental Expenses

Notes: The figure shows the dynamic effects of property taxes on business rental expenses. The dependent variable is rent costs scaled by its pre-reform value, implying the coefficients can be interpreted as a percent change after multiplying by 100. The horizontal axis shows the number of years since the policy. Each dot represents the point estimate of the treatment effect for that year, with the reference category set at -1. The vertical bands indicate the corresponding 90 percent confidence intervals. The regression specification includes fixed effects for sector, pre-reform size, and accounting month. The coefficients are estimated using a staggered Difference-in-Differences approach, following Callaway and Sant'Anna (2021), with not-yet treated units used as the comparison group. Appendix Figure A.4 shows the event-study plot using never treated units as the comparison group. Standard errors are clustered at the firm level.

Implied Pass-Through Rate:— To interpret the economic magnitude of the reform's effects on rental expenses, I compute an implied pass-through rate using the reduced-form estimate. Because property taxes are levied at the property level and the data do not link tenant firms to landlords, I do not observe firm-specific property tax liabilities. Moreover, multiple firms may operate within the same property. As a result, it is not possible to construct a firm-level measure of tax exposure or pass-through. Given these limitations, I estimate pass-through at the citywide level. Conceptually, this approach treats the reform as an aggregate shock to the local cost of supplying commercial property and asks what

fraction of the resulting increase in potential property tax revenue is transmitted to firms through higher rents. I operationalize this by scaling the reduced-form increase in rents paid by firms by the aggregate increase in citywide property tax liabilities induced by the reform. Formally, I define the citywide pass-through rate as: $\frac{\hat{\beta} * \bar{R} * N}{\Delta T^{\text{City}}}$, where $\bar{R}$ denotes the average pre-reform rent expense, N is the total number of firms operating in the city prior to the reform, and ΔT^{City} is the reform-induced change in citywide property tax liabilities. This measure captures the share of the aggregate property tax increase that is ultimately borne by firms through higher rents. Importantly, it should not be interpreted as a structural, firm-level pass-through rate. Instead, it represents an incidence measure that aggregates across heterogeneous exposure to the reform, endogenous firm responses, and any general equilibrium effects operating through the local rental market.

Results in Table 2 (second row from the bottom) indicate that a UGX 1 increase in potential citywide property tax liabilities is associated with a UGX 0.42 increase in total rents paid by firms. Interpreted as an incidence measure at the city level, this suggests that approximately 42 percent of the aggregate commercial property tax burden is transmitted to firms through higher rents. Given that only about 65 percent of potential commercial property tax liabilities are ultimately collected, this estimate implies that a UGX 1 increase in collected citywide property tax revenue is associated with a UGX 0.65 increase in total rents paid by firms.

While not directly comparable to property-level pass-through estimates in the housing literature, the citywide incidence measure estimated here is surprisingly similar in magnitude to recent studies of property tax pass-through in rental housing markets. These studies typically estimate pass-through at the property or unit level and, in some cases, embed the analysis in a general equilibrium framework that allows for endogenous adjustment in rents, occupancy, and housing supply, finding pass-through rates ranging from \$0.50 to \$1.39 per \$1 of tax (e.g., Baker 2024; Watson and Ziv 2024; Löffler and Siegloch 2021; Rolheiser 2019). By contrast, my estimate captures a reduced form incidence measure at the city level, aggregating across heterogeneous properties, landlords, and tenant firms, and abstracts from equilibrium reallocation across locations. Despite these differences in level of aggregation, empirical design, and the study context, the similarity in magnitudes is informative and suggests that a substantial share of the commercial property tax burden is transmitted to firms through higher rents.

My results are consistent with the "old view" of property tax incidence, which asserts that the portion of the tax falling on land is borne entirely by owners, since the supply of land is perfectly inelastic, while the portion falling on improvements— i.e., structures— can be shifted to final consumers and occupants, since the supply of capital to the con-

struction industry is relatively elastic (Oates and Fischel 2016). In my setting, it is plausible to assume that the entire tax falls on improvements rather than on land itself, since idle land is exempt.

Table 2: Effects of Property Tax on Business Rental Expenses

	(1)	(2)	(3)	(4)
Average Effect	0.069**	0.096***	0.097***	0.090***
	(0.030)	(0.031)	(0.031)	(0.031)
Sector of Operation	N	Y	Y	Y
Pre-reform Size	N	N	Y	Y
Other Controls	N	N	N	Y
Pre-reform Mean	35.46	35.46	35.46	35.46
Implied Pass-Through Rate	0.322	0.447	0.452	0.423
Observations	30798	30798	30798	30798

Note: The table presents average effects of the reform on business rental expenses. The dependent variable is rent costs scaled by its pre-reform value, implying the coefficients can be interpreted as a percent change after multiplying by 100. Each column reports estimates from a different specification with an alternative set of controls. Under "Other Controls," I include dummies for accounting month, and whether the firm filed returns on time, all measured in the pre-reform period. The bottom row presents citywide pass-through calculated from the expression: $\frac{\hat{\beta} * \bar{R} * N}{\Delta T^{\text{City}}}$, where $\bar{R}$ denotes the average pre-reform rent expense, N is the total number of firms operating in the city prior to the reform, and ΔT^{City} is the reform-induced change in citywide property tax liabilities. The coefficients are estimated using a staggered Difference-in-Differences approach following Callaway and Sant'Anna (2021), using not-yet-treated units as the comparison group. Standard errors are clustered at the firm level. Pre-reform mean is measured in millions of Uganda Shillings.

V.B. Occupancy Status: Own or Lease?

Since the incidence of property taxes depends critically on occupancy status, higher tax liabilities may prompt firms to reconsider whether to own or lease their operating space. Owner-occupiers facing higher property tax bills may find leasing comparatively more attractive, while tenants, who bear the tax indirectly through landlord pass-through, face weaker incentives to switch. I examine this occupancy margin directly using three outcome variables that together characterize the full distribution of ownership arrangements.

I first examine whether the reform induced complete switches between full ownership and full tenancy. The outcome takes the value one if the firm reports positive land and building values with zero rent expenses, and zero if the firm reports positive rent expenses with zero land and building values, excluding firms that report positive values in both categories. Panel A of Appendix Figure A.2 shows that the reform had no discernible effect on full occupancy status: firms did not wholesale switch from owning to

renting, or vice versa. This is perhaps unsurprising; complete divestiture or acquisition of property is a large, lumpy, and costly decision that is unlikely to be triggered by a tax reform within a short horizon, particularly given the illiquidity of commercial real estate markets in developing countries.

The more revealing margin is at the intensive edge of occupancy arrangements. A non-trivial share of firms in the data simultaneously report positive values in both land and buildings and rent expenses, reflecting partial occupancy arrangements whereby firms own their primary premises but lease supplementary space, or vice versa. I exploit this feature to construct two additional outcome variables that capture transitions into partial arrangements from either direction. Panel B of Appendix Figure A.2 shows that the reform significantly increased the likelihood of fully-owning firms moving into partial arrangements; reporting both owned and leased space, by 6.2 percentage points against a pre-reform mean of 41 percent. This suggests that owner-occupiers, facing higher property tax liabilities, respond at the margin by leasing additional or replacement space rather than divesting owned property outright, a form of gradual portfolio rebalancing that avoids the transaction costs of full divestiture. Panel C shows no corresponding effect for full renters transitioning into partial arrangements, consistent with the weaker incentive for tenants to acquire property in response to a tax that is only partially passed through to them.

Taken together, these findings reveal that property tax reform reshapes occupancy decisions primarily through an intensive adjustment margin rather than complete tenure switching. Owner-occupiers do not sell their properties, but they do begin to incorporate leased space into their operations, gradually diluting their exposure to property tax liabilities.

V.C. Theoretical Framework

The result that higher property tax liabilities lead to an increase in business rent expenses contrasts with the predictions of neoclassical theory, which suggests little to no pass-through of property taxes (Mieszkowski 1972; Zodrow 2001). Underlying the neoclassical prediction is the idea that demand for space in a location is elastic, since other jurisdictions offer comparable commercial units, while supply is inelastic. In that setting, the property tax falls on the immobile factor and landlords absorb the entire burden because any attempt to raise rents would drive tenants to competing locations.

A necessary condition for this result, however, is that the property tax is a cost of *owning* the property, not a cost of *renting it out*. If the tax is enforced universally, the landlord owes the same liability whether the property is occupied or vacant, and the tax

is effectively a lump sum that cannot influence the landlord's pricing decision. But if the tax is enforced selectively — collected only when the property is occupied and generating observable economic activity — then renting out the property triggers the tax liability. In this case, the tax functions as a marginal cost of supplying space to the rental market, and standard incidence logic predicts positive pass-through to tenants.

To rationalize my findings, I develop a simple framework that predicts non-zero pass-through and that does not depend on the elasticities of demand and supply in the standard sense. The framework embeds a key institutional feature of property tax enforcement in developing countries: *selective enforcement*, where the probability of enforcing, and therefore collecting, property taxes is lower on vacant properties than on occupied ones. This pattern arises naturally in settings where tax administrations have limited capacity and prioritize enforcement on properties that are visibly in use and generating income (Franzsen and McCluskey 2017). When administrative resources are scarce, targeting occupied properties is a cost-effective strategy: occupied properties are easier to identify, their owners are easier to locate, and the revenue yield per enforcement action is higher because an active tenant signals ability to pay. Consistent with the pattern of selective enforcement documented above, Ahabwe et al. (2020) notes that demand letters in Kampala are disproportionately directed toward high-value property owners, reflecting the administrative reality that limited enforcement resources incentivize tax authorities to prioritize collections where the potential yield is greatest.

Setup:— Consider a landlord who owns one unit of commercial space and must choose the rent R to charge. If the unit is occupied at rent R, the landlord earns rental income R and pays the property tax liability $T = \tau V$ with certainty, yielding a payoff of $R - T$.[7] If the unit remains vacant, the landlord earns no rental income but may still face the tax: the tax is enforced on the vacant unit with probability $p_e \leq 1$, yielding an expected payoff of $-p_e T$. [8]

The probability that the unit is occupied depends on the rent charged. Let $q(R)$ denote the probability that the unit is filled at rent R, with $q'(R) < 0$ (higher rents reduce the likelihood of finding a tenant) and $q(0) \leq 1$. Under the assumption of a unit mass of potential tenant businesses, $q(R)$ also represents the aggregate demand for commercial space, i.e., the share of businesses willing to pay rent R rather than take their outside option.[9]

[7] τ is the property tax rate and V is the assessed value.

[8] In a setting with imperfect compliance, even when the unit is occupied, as is common in many developing countries, p_e can be interpreted as the relative probability of compliance when the unit is vacant compared to when it is occupied. If we assume landlords are more likely to comply when the unit is occupied compared to when it's vacant, then $p_e < 1$.

[9] This equivalence holds when each potential tenant independently decides whether to rent the unit. If

Landlord's Maximization Problem:— The landlord sets rent to maximize his expected utility:

$$\max_{R} \; q(R)\,(R-T) + (1-q(R))\,(-p_e T) \tag{4}$$

The first term is the expected payoff if the unit is filled: the landlord earns $R-T$ with probability $q(R)$. The second term is the expected payoff if the unit remains vacant: the landlord pays $p_e T$ in expected tax with probability $1-q(R)$.

Expanding (4) and dropping the terms that do not depend on R, the landlord effectively maximizes:

$$\max_{R} \; q(R)\,[R-(1-p_e)T] \tag{5}$$

This has a clear economic interpretation: the landlord behaves as if facing a per-unit cost of $(1-p_e)T$ for each occupied unit. This cost is the additional expected tax the landlord incurs by having the unit occupied rather than vacant. Under universal enforcement ($p_e = 1$), this cost is zero— the landlord pays T regardless of occupancy, so occupancy triggers no additional liability. Under fully selective enforcement ($p_e = 0$), the cost is the full tax T— each occupied unit triggers the entire tax bill.

The first-order condition of (5) is:

$$q'(R)\,[R-(1-p_e)T] + q(R) = 0 \tag{6}$$

Solving for R:

$$\boxed{R^* = (1-p_e)T - \frac{q(R^*)}{q'(R^*)}} \tag{7}$$

The optimal rent has two components. The first, $(1-p_e)T$, is the effective marginal cost of occupancy—the portion of the tax that the landlord can avoid by leaving the unit vacant. The second, $-q(R^*)/q'(R^*)$, is the standard markup that depends on the shape of the demand curve. Under universal enforcement ($p_e = 1$), the first term vanishes and the optimal rent depends only on demand conditions, not on the tax, recovering the neoclassical result.

Characterizing the Pass-through Rate:— Defining $c \equiv (1-p_e)T$ as the effective marginal cost of occupancy, the landlord's first-order condition implicitly defines the optimal rent R^* as a function of the effective cost c, and hence of T. Define the function:

$$G(R,c) \equiv q'(R)(R-c) + q(R) = 0 \tag{8}$$

tenants have heterogeneous reservation values for the location, $q(R)$ is the fraction whose reservation value exceeds R.

By the implicit function theorem we have:

$$\frac{dR^*}{dc} = -\frac{-q'(R^*)}{q''(R^*)(R^* - c) + 2q'(R^*)} = \frac{q'(R^*)}{q''(R^*)(R^* - c) + 2q'(R^*)} \tag{9}$$

From the first-order condition (6), we can substitute $R^* - c = -q(R^*)/q'(R^*)$, we can write (9) as:

$$\frac{dR^*}{dc} = \frac{1}{2 - \dfrac{q(R^*)\, q''(R^*)}{[q'(R^*)]^2}} \tag{10}$$

Since $c = (1 - p_e)T$ and hence $dc/dT = (1 - p_e)$, the pass-through rate with respect to the tax liability is:

$$\boxed{\rho = \frac{dR^*}{dT} = \frac{dR^*}{dc} \cdot \frac{dc}{dT} = \frac{1 - p_e}{2 - \theta}}, \tag{11}$$

where $\theta \equiv \frac{q(R^*)\, q''(R^*)}{[q'(R^*)]^2}$ is the demand curvature parameter introduced in the tax incidence literature by Weyl and Fabinger (2013). Note that the pass-through formula (11) requires $\theta < 2$ for $\rho \geq 0$. This is not an additional assumption but rather the second-order condition for the landlord's maximization problem, which must hold for the optimal rent R^* to be a well-defined interior maximum (See Appendix, Section II.A for derivation).

The parameter θ determines the distribution of the burden for a given $p_e \in (0, 1)$. The pass-through rate is increasing the θ for any given p_e. Equation 11 also allows for the possibility of over-shifting if $\theta > 1$ (strongly log-convex demand) and p_e is sufficiently small. If we assume a simple, linear demand, $q(R) = q_0 - q_1 R$, we have $q'' = 0$ and therefore $\theta = 0$. The pass-through rate is $\rho = (1 - p_e)/2$. Under thus special case, the burden of property taxes is equally shared between the landlord and the tenant if $p_e = 0$, the case in which there is no enforcement on vacant units. In this case, the landlord bears more than half the burden if $\theta < 0$ (log-concave demand), and the tenant bears more than half if $\theta > 0$ (log-convex demand).

VI. EFFECTS ON BUSINESS ACTIVITY

VI.A. Revenues, Profits, and Investment

Property tax liabilities raise occupancy-related costs — directly for owner-occupiers through higher tax bills, and indirectly for tenants through landlord pass-through, as documented in the previous section. These cost increases may propagate into broader firm performance, compressing profit margins, crowding out investment, and ultimately

affecting revenues. To examine these effects, I estimate the reform's impact on log revenues, profits normalized by pre-reform revenues, and investment rates, using a balanced panel of active firms.[10]

Figure 5 presents the results. The absence of differential pre-reform trends across all three outcomes among firms in treated relative to untreated areas lends further support to the parallel trends assumption underlying the identification strategy. Following the reform, revenues show no statistically significant response on average, while profits and investment decline markedly. Firms' profits fall by 2.4 percent relative to pre-reform revenues, and investment rates decline by 2.7 percentage points. The profit result is consistent with firms absorbing part of the burden associated with property taxes. Although this is not purely a business tax, this pattern aligns with evidence on local business tax incidence in settings with competitive product markets (Suárez Serrato and Zidar 2016). The investment decline suggests that the cost shock not only squeezes current profitability but also depresses capital accumulation, potentially generating persistent effects on firm productivity and growth. Appendix Figure A.5 corroborates these patterns, showing differential declines in value added, capital stock, and investment at the extensive margin. Average effects are reported in Appendix Table A.1.[11]

These average effects, however, mask substantial heterogeneity by firm size. Appendix Figure A.3 shows that the profit and revenue declines are concentrated among small firms, while large firms experience revenue gains with profits and investment left largely unchanged. This divergence is consistent with large firms exercising greater market power, enabling them to pass a share of the burden onto consumers through higher output prices. If the demand for their products is sufficiently inelastic, such price adjustments can raise revenues even as costs increase. Small firms, by contrast, operate in more competitive environments or face more elastic demand, leaving them unable to adjust prices and therefore forced to absorb the cost increase directly into margins. This size-dependent heterogeneity in pass-through capacity has important implications for the distributional incidence of commercial property taxation: the burden falls disproportionately on smaller, less financially resilient firms, while larger incumbents are relatively insulated, and may even strengthen their competitive position relative to smaller rivals as a result.

[10] Active firms are defined as those reporting non-zero turnover in at least one period.

[11] The investment rate is defined as the annual change in net fixed assets scaled by lagged fixed assets.

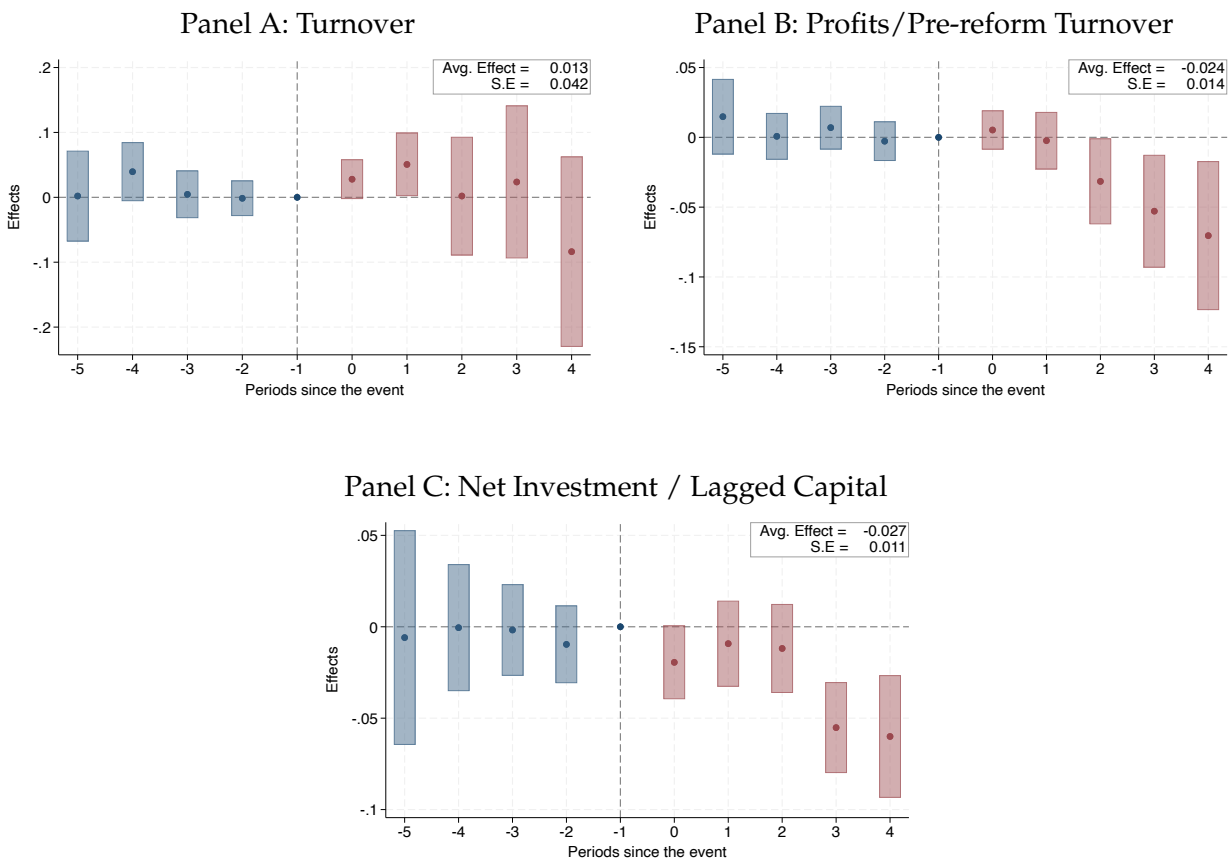


Figure 5: Effects of the Reform on Revenues, Profits, and Investment

Notes: The Figure presents the dynamic effects of the reform on revenues, profits, and investment. Revenues are transformed in logs. Profits are defined as earnings before interest and taxes (EBIT), scaled by pre-reform revenues. Investment rate is defined as the annual change in net fixed assets, scaled by lagged fixed assets. The horizontal axis indicates the number of years since the reform. Each dot represents the point estimate of the treatment effect for that year, with the reference category set at -1. The vertical bands indicate the corresponding 90 percent confidence intervals. The regression specification includes fixed effects for sector and pre-reform size bins. The coefficients are estimated using a staggered Difference-in-Differences approach, following Callaway and Sant'Anna (2021), with not-yet treated units used as the comparison group. Standard errors are clustered at the firm level.

VI.B. Firm Survival

Beyond their effects on revenues, profits, and investment, property taxes may shape the very existence of firms in the local economy. A firm's decision to continue operating depends on whether expected future profits exceed the opportunity cost of remaining in business, a margin that a permanent increase in property tax liabilities can directly erode. For owner-occupiers, higher property taxes raise the fixed cost of maintaining a physical presence, compressing the profit floor below which exit becomes optimal. For tenants, landlord pass-through translates the tax increase into higher rents, with similar consequences for the exit margin. Critically, unlike variable costs that adjust with output, property taxes are largely fixed obligations that do not flex with firm performance, making them particularly burdensome during periods of low revenue. The survival margin is therefore not merely a reduced-form outcome of interest, but a key channel through which property taxation shapes the size and the composition of the local firm economy. In this section, I investigate whether the reform affected the probability of firm survival.

Constructing the Survival Outcome:—Analyzing firm survival in firm-level panel data is challenging because firm exits inherently renders the sample endogenous to the reform. To address this, I fix the cohort to firms present in 2015/16 (the year prior to the reform), with positive turnover, regardless of whether they operated in other years.[12] I then construct a balanced panel by expanding each firm to all sample years and assigning zeros for turnover, sales, and related variables in the years the firm did not operate. The outcome variable is then defined as an indicator equal to 1 if a firm has positive turnover in year t and 0 otherwise. A similar approach was used by Saez et al. (2019) when analyzing whether payroll taxes affects firm survival in Sweden. By construction, the mean of the outcome variable is one in the reference year; it can be below one before 2016 because some firms had not yet entered, and below one after 2016 because some firms exit following the reform.

However, as highlighted by Saez et al. (2019), using this approach to estimate the effects on firm survival requires reweighting firms to align pre-reform distributions along key dimensions across different groups, since unweighted estimates can easily reflect compositional differences rather than the causal effects. For example, if younger firms, which tend to have lower baseline survival rates, are unevenly distributed across treated and untreated locations, unweighted estimates may simply capture age-related compositional differences rather than the effect of the tax reform itself. To address this, I reweight firms using the nonparametric reweighting method of DiNardo et al. (1996), following the

[12] As a robustness check, I repeat the this procedure using the cohort of firms present in the first year of the sample (with positive turnover)

implementation in Yagan (2015). Observations are first weighted by average pre-reform turnover, so that estimates reflect economic scale and larger firms contribute proportionally more to the parameter estimates. I then reweight the sample across size-industry bins to ensure that treated and control groups are balanced along these dimensions in the pre-reform period, so that time-varying shocks to particular size or industry cells do not differentially influence the survival estimates.

Effects on Firm Survival:—Figure 6 shows the effects of the reform on firm survival using the reweighted sample constructed following the procedure described above. In the pre-reform period, there is no evidence of differential trends in survival probabilities between firms located in areas that experienced increases in property tax liabilities and those in areas that did not, as all pre-reform coefficients are statistically indistinguishable from zero. This finding supports the underlying identifying assumption that, absent the reform, survival probabilities for treated and untreated firms would have evolved similarly, allowing post-reform effects to be interpreted as causal effects of the reform.

Following the reform, there is a differential decline in survival probabilities for firms located in areas that experienced an increase in property tax liabilities relative to firms in locations that did not experience the reform, implying the reform increased firm exit. The negative effects start to manifest after one year of the reform, grows in the second year before it stabilizes at proximately 5 percentage points and persists up to the end of the sample. The corresponding average effects are presented in Table 3. Column 4 shows the preferred specification which includes a full set of controls. Comparing firms in locations that experienced increase in property tax liability versus those in other locations, the reform led to a 3.2 percent decline in survival probabilities with standard error of 1.1 percent. Scaling this effect by the aggregate increase in potential property tax liabilities implies that a $1 million increase in aggregate property tax liabilities leads to approximately 25 additional firm exits (see bottom row of Table 3).

A key concern with this approach to measuring firm survival is that it may be confounded by evasion responses. Firms may respond to the higher costs associated with increased property tax liabilities by ceasing to file income tax returns, in which case this approach would misclassify them as exits. I implement several strategies to deal with this concern. First, I use a cross-tax validation strategy to identify firms' operational footprints. Using employee-level tax return data in combination with CIT data, I classify evasion as cases where the firm disappears from the CIT but still has employees associated with its tax identification. Accordingly, I classify a case as exit if the firm disappears from the CIT and has no footprints in the employee tax returns, i.e., both taxes go silent. Appendix Figure A.6 shows that my estimates are unlikely to be driven by an evasion

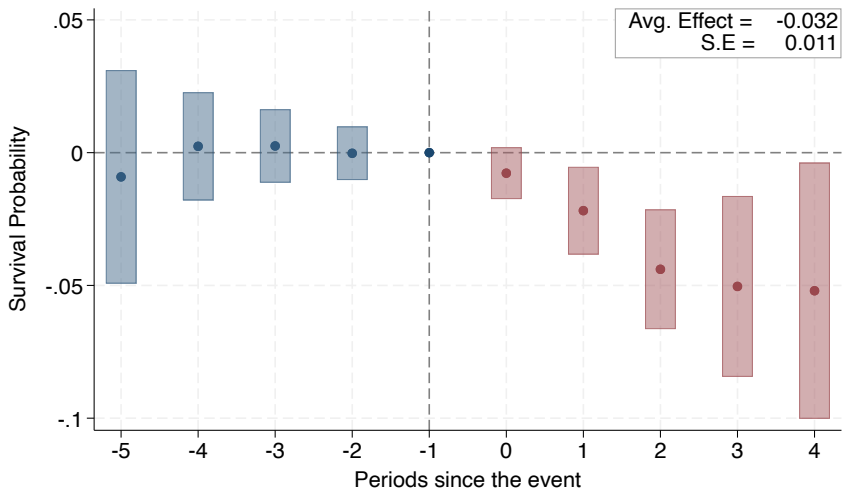


Figure 6: Effects of Property Tax Enforcement on Firm Survival

Notes: The figure presents the dynamic effects of the reform on firm survival. The outcome variable is an indicator equal to one if the firm is operating at time t, zero otherwise, implying the coefficients can be interpreted as percentage point change in survival probability. The horizontal axis indicates the number of years since the reform. Each dot represents the point estimate of the treatment effect for that year, with the reference category set at -1. The vertical bands indicate the corresponding 90 percent confidence intervals. The regression specification includes fixed effects for sector and pre-reform size bins. The coefficients are estimated using a staggered Difference-in-Differences approach, following Callaway and Sant'Anna (2021), with not-yet treated units used as the comparison group. Standard errors are clustered at the firm level.

response. A likely criticism of this validation approach is that firms may collude with workers to evade both taxes. However, the timing mismatch between the two tax returns makes such collusion practically difficult. Employee tax returns are filed monthly, whereas corporate income tax (CIT) is filed annually. Since I classify a case as evasion even if the firm's identifier appears in only one month, for collusion to occur, the firm would have to decide to evade at the start of the income year, that is, before the realization of output. Moreover, workers would need to be compensated for agreeing to collude, as doing so would entail forfeiting other benefits such as social security. This creates an additional constraint on the firm's ability to engage in coordinated evasion. To my knowledge, this paper is the first to cleanly separate survival effects from evasion response using the cross-tax referencing.

Other Robustness Checks:— I conduct several robustness checks to assess the sensitivity of the survival effects, with results reported in the Appendix. First, I redefine exit as the absence of a corporate income tax (CIT) return for at least two consecutive years, thereby excluding firms that temporarily disappear for one year and subsequently re-enter. Second, I impose a stricter definition of exit that requires no CIT return in year t and no record in employee tax returns in both years t and $t+1$. As shown in Figure

Table 3: Effects of Property Tax Enforcement on Firm Survival

	(1)	(2)	(3)	(4)
ATE	-0.036***	-0.029***	-0.029**	-0.032***
	(0.011)	(0.011)	(0.011)	(0.011)
Sector of Operation	No	Yes	Yes	Yes
Pre-reform Size	No	No	Yes	Yes
Other Controls	No	No	No	Yes
Observations	130864	123864	123864	123864
Exits per $1 Million	28.35	22.90	22.71	25.34

Note: The table presents average effects of the reform on firm survival probabilities. The outcome variable is an indicator equal to one if the firm is operating at time t, zero otherwise. Because the mean of the dependent variable is one in pre-reform year by construction, the coefficient estimates can be interpreted as percent change in survival probability. Each column reports estimates from a different specification with an alternative set of controls. Under "Other Controls," I include dummies for accounting month, whether the firm reported owning land and buildings, and whether the firm filed returns on time, all measured in the pre-reform period. The bottom row reports the citywide reform-induced firm exits per additional $1 million in aggregate property tax liability, calculated as $\frac{\hat{\beta} \cdot N}{\Delta T^{\text{City}}}$, where N denotes the total number of active firms in the city prior to the reform and ΔT^{City} is the reform-induced change in citywide property tax liability. The coefficients are estimated using a staggered Difference-in-Differences approach following Callaway and Sant'Anna (2021), with not-yet treated units used as the comparison group. Standard errors are clustered at the firm level.

A.7, the estimated effects remain stable under both definitions. Third, I reconstruct the survival outcome following the same procedure as before but restrict the sample to the cohort of firms that were present and reported positive turnover in the first year of the sample. Figure A.8 shows that the results are consistent with those obtained using the baseline sample. As a final robustness check, I assess whether the findings are sensitive to the choice of comparison group in the Callaway and Sant'Anna (2021) difference-in-differences design. Figure A.9 shows that the results remain robust when using never-treated firms as the comparison group rather than not-yet-treated firms.

VI.C. Labor Market Outcomes

Effects on Formal Employment:— The preceding results on firm survival have direct implications for employment. Firm exits lead to job losses, and surviving firms may also adjust employment in response to the reform. I begin by analyzing employment effects using the broader sample from section VI.B, which includes exiting firms, coding employment as zero following exit. I also examine whether continuously active firms adjust employment in response to the higher costs associated with increased property tax liabilities. For this part, I focus on a balanced panel of firms that remain in operation; that is,

firms with positive turnover in every year of the sample.

Because I observe only formal employment, which can be zero for some firms even prior to exit, and because employment is coded as zero after firm exit, I do not transform the outcome using logarithms, as this would complicate the interpretation of coefficients as percentage effects, as highlighted by Chen and Roth (2024). Instead, I normalize employment relative to its pre-reform level (2015/16), setting the normalized outcome to zero when employment is zero in both year t and the pre-reform year. I then winsorize the normalized outcome at the 95th percentile. Normalizing outcomes by pre-reform values is common in the literature (e.g., Saez et al. 2019; Harju et al. 2022) and facilitates an economically meaningful interpretation of the regression coefficients in percent changes or percentage points. As a robustness check, I re-estimate the employment effects using alternative transformations of the outcome variable: log(1+employment), the inverse hyperbolic sine transformation, and levels. Appendix Figure A.12 shows that the event-study estimates and average effects are consistent across all three transformations, confirming that the baseline results are not sensitive to the choice of outcome transformation.

Figure 7 presents the event-study estimates and the corresponding 95% confidence intervals for the impact of the reform on formal employment, estimated following Callaway and Sant'Anna (2021). Prior to the reform, formal employment in firms located in treated areas follows similar trends to those in untreated areas. Following the reform, employment declines differentially among firms in treated areas relative to those in untreated areas. The negative effect emerges in the reform year, grows for the next three years before it stabilizes at proximately 7 percent and persists up to the end of the sample.

The average effects of the reform on employment are reported in Table 4. On average, employment in firms located in treated areas declined by 4.6 percent relative to those in untreated areas (Column 3). Appendix Figure A.11 shows that employment declines even when the sample is restricted to firms active throughout the entire sample period, indicating that the aggregate employment effect is not solely driven by firm exit but also reflects deliberate workforce reductions among surviving firms. This within-firm adjustment among survivors signal that even firms that weather the reform do so partly by contracting their workforce in response to the higher cost burden. Together, these two margins suggest that the employment consequences of property taxation extend beyond the extensive margin of firm survival documented in the previous section.

To assess the economic magnitude of these effects, I translate the estimated percent decline in employment into an aggregate citywide measure. Because, I do not observe firm-specific property tax liabilities for reasons highlighted in section V.A, I can not com-

pute firm-level employment elasticities with respect to property tax liabilities. Instead, I aggregate the reduced form employment effects across firms and scale them by the aggregate increase in potential property tax liabilities induced by the reform. The reduced form estimate therefore implies that a $1 million increase in aggregate property tax liabilities is associated with approximately 170 fewer jobs citywide (see bottom row, Table 4, column 4).

While not directly comparable due to differences in the type taxes studied, the negative effects on employment here are consistent with previous studies on payroll taxes (Saez et al. 2019; Egebark and Kaunitz 2018) and corporate taxes (Ljungqvist and Smolyansky 2014; Giroud and Rauh 2019).

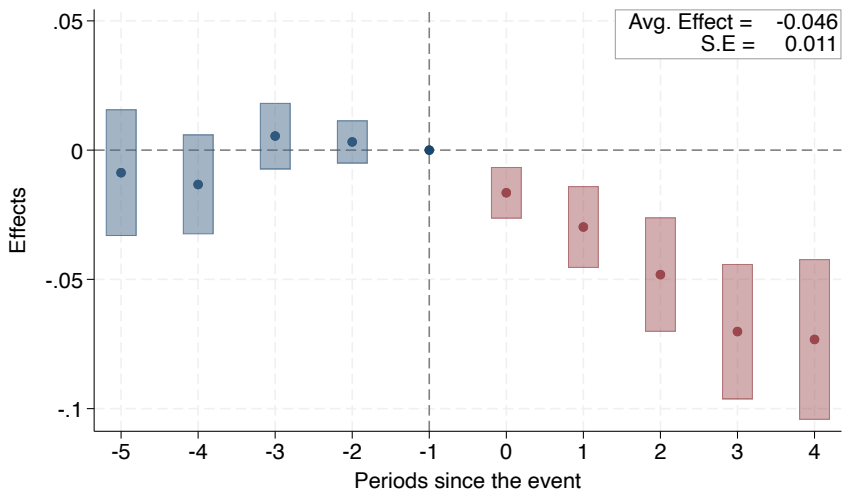


Figure 7: Effects of Property Tax Enforcement on Formal Employment

Notes: The figure presents the dynamic effects of the reform on employment. The dependent variable is normalized by its pre-reform value, implying the coefficients can be interpreted as a percent changes. Each dot denotes the point estimate of the treatment effect for that year, with the reference period set at -1. The vertical bands indicate the corresponding 95 percent confidence intervals. The regression specification includes fixed effects for sector and pre-reform size. Standard errors are clustered at the firm level.

Effects on Firms' Wage Bill:— Panel A of Figure 8 documents a differential decline in the total wage bill of treated relative to untreated firms following the reform, with an average effect of 5.5 percent. This reduction operates through two complementary channels: the mechanical contraction of the wage bill arising from the decline in formal employment documented above, and deliberate wage adjustments by firms that retain their workforce but compress compensation in response to higher cost burdens. To the extent that both channels are operative implies that the labor cost consequences of the reform extend beyond headcount reductions alone.

Table 4: Effects of Property Tax Enforcement on Formal Employment

	(1)	(2)	(3)
ATE	-0.052***	-0.050***	-0.046***
	(0.010)	(0.012)	(0.011)
Industry-Size	N	Y	Y
Other Controls	N	N	Y
Pre-reform Mean	6.193	6.193	6.193
Observations	123921	118584	118584
Jobs Lost per $1 Million	192	185	170

Note: The table presents the average effects of property taxes on the employment, normalized by its pre-reform value. Each column reports estimates from a different specification with an alternative set of control. Under "Other Controls," I include dummies for accounting month, whether the firm reported owning land and buildings, and whether the firm filed returns on time, all measured in the pre-reform period. The bottom row reports the citywide reform-induced job loss per additional $1 million in aggregate commercial property tax liability, calculated as $\frac{\hat{\beta}\cdot\bar{E}\cdot N}{\Delta T^{\text{City}}}$, where $\hat{\beta}$ is the reduced form coefficient, $\bar{E}$ denotes average pre-form employment, N denotes the total number of active firms in the city prior to the reform and ΔT^{City} is the reform-induced change in citywide commercial property tax liability.

Firms facing higher costs associated with increase in property tax liabilities may have an incentive to shift workers off formal payrolls—reducing their social security contributions and other payroll-linked obligations—while retaining their productive services through informal contracts, a marginal adjustment that would be invisible to analyses relying solely on formal employment records. To analyze this margin, I construct a measure of informal wages, as the difference between the wage bill reported on firms' corporate income tax returns and the total wage bill derived from employee-level returns. I normalize the outcome variable by the CIT wage bill, so that coefficients are interpretable as percentage-point changes.[13]

Panel B of Figure 8 that the share of informal wages increased by 3.3 percentage points against a pre-reform mean of 10.4 percent. This suggests that a non-trivial share of the observed decline in formal employment may reflect a reallocation of labor toward informal arrangements rather than outright job destruction, implying that the aggregate employment estimates above overstate true job losses to the extent that some workers transition into informal roles rather than unemployment. Beyond the measurement implication, this informalization margin points to a broader distributional consequence of property taxation in contexts where informal labor markets are prevalent: the tax burden is partly shifted onto workers in the form of reduced labor protections and benefits associated with

[13]The underlying assumption is the firms report truthfully on their wage bill on the corporate income tax returns.

formal employment.

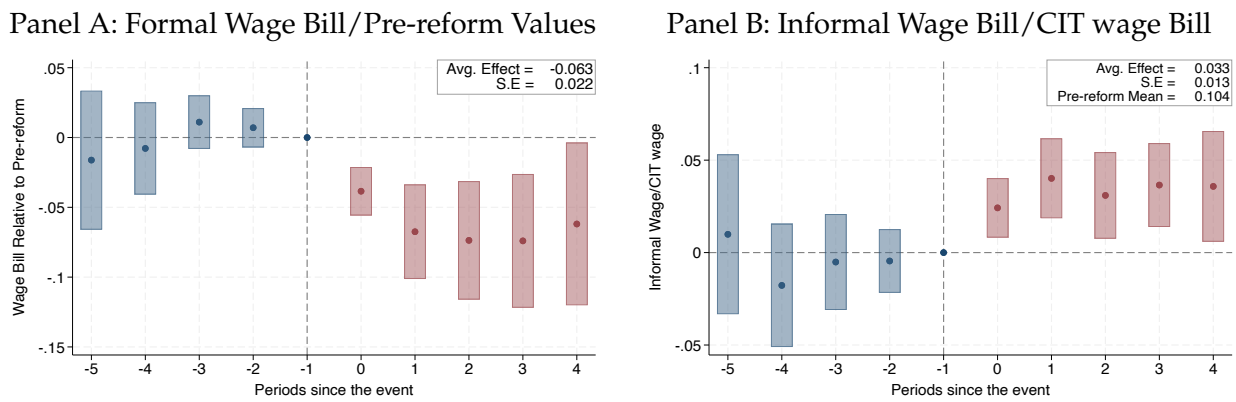


Figure 8: Property Taxes and Firms' Wage Bill

Notes: The figure presents the dynamic effects of the reform on firms' total wage bill. In Panel A, the dependent variable is total wage bill, scaled by its pre-reform value, i.e., $\frac{W_{i,t}}{W_{i,0}}$, where $W_{i,t}$ denotes firm i's formal wage bill at time t and $W_{i,0}$ denotes the pre-reform period wage bill. In Panel B, the dependent variable is informal wage, defined as $\frac{\text{CIT wage bill - Total age bill from employee returns}}{\text{CIT wage bill}}$. This means coefficients can be interpreted as a percent changes. The horizontal axis reports the number of years relative to the reform implementation. Each dot represents the estimated treatment effect for a given year, with year -1 as the reference category. The vertical bands denote the corresponding 95 percent confidence intervals. The regression specification includes fixed effects for sector, pre-reform size, and accounting month. Standard errors are clustered at the firm level.

A potential concern with the wage informality analysis is that the reform may incentivize firms to inflate their reported wage bill on corporate income tax returns in order to reduce taxable income, which would generate a spurious wedge between CIT-reported and employee-level wages. If firms were systematically over-reporting labor costs in response to the reform, one would expect a broad-based inflation of other deductible expense categories as well. Figure **??** provides evidence against this interpretation: telephone, computer, machine repair, and other unclassified operating expenses all decline following the reform, with no differential pre-reform trends. Rather than reflecting strategic over-reporting, this pattern is consistent with firms cutting operating costs broadly in response to higher costs associated with increase in property tax liabilities, corroborating the interpretation that the wedge between CIT-reported and employee-level wages reflects genuine labor informalization rather than tax-motivated accounting manipulation.

VII. DISTRIBUTIONAL CONSEQUENCES

The aggregate employment effects documented above mask important heterogeneity in how the burden of the reform is distributed across workers. In particular, if firms

respond to the reform by adjusting employment at the bottom of the wage distribution—cutting low-skill, low-wage workers first—the reform may disproportionately harm the most economically vulnerable workers, even when its aggregate employment effect appears moderate. As with other business taxes, the distributional consequences of property taxation thus depend critically on which workers bear the brunt of firms' adjustment margins (e.g., Risch 2024; Fuest et al. 2018).

To identify which types of workers are most affected by the reform, I pursue two complementary firm-level analyses.[14] First, I examine changes in employment composition by tracking how the share of workers in each pre-reform wage quartile, relative to pre-reform total employment, evolves around the reform. Formally, the outcome variable is $\frac{E_{q,t}}{E_0}$, where $E_{b,t}$ denotes the number of workers in quartile q at time t and E_0 denotes pre-reform total employment.[15] Second, I analyze how the within-firm wage distribution evolves in treated relative to untreated firms following the reform. In both analyses, I restrict the sample to firms that employed at least five workers in the pre-reform period and remained active throughout the sample period, to ensure adequate support across the wage distribution. All estimates follow the staggered difference-in-differences approach of Callaway and Sant'Anna (2021).

Figure 9 presents the effects of the reform on employment composition across the pre-reform wage distribution. The reform led to a differential decline in the share of workers in the top quartile among treated relative to untreated firms, accompanied by a corresponding increase in the share of workers in the 50th–75th percentile. The absence of pre-reform differential trends in Panel A supports a causal interpretation of these patterns. On average, the share of workers in the top quartile declined by 2.8 percentage points, while the share in the 550th–75th percentile increased by 2.2 percentage points (Panel B). Effects on the bottom quartile and the 25th–50th percentile are statistically insignificant, though the point estimates are positive.

Two mechanisms could account for this pattern. First, firms may substitute high-earning workers with lower-wage workers of similar productivity, reducing their wage bill without proportionally contracting their workforce. Second, wages at the top of the distribution may have fallen more sharply than those at the bottom, mechanically shifting workers from the top quartile into the third quartile without any change in employment composition per se.

In the second analysis, I investigate the wage channel by examining how the within-

[14] I conduct the analysis at the firm level rather than the worker level, as workers are uniquely identified only at the firm level in the data.

[15] The focus on the quartiles reflects sample size limitations.

firm wage distribution evolves before and after the reform. Specifically, I track the 20th, 40th, 60th, and 80th percentiles of the within-firm wage distribution in treated relative to untreated firms, scaling each percentile by the pre-reform total formal wage bill to allow interpretation of the estimates as percentage-point changes relative to the pre-reform wage bill. Figure A.13 shows that while coefficient estimates are negative across the entire distribution, the effects are stronger at the top: the 60th percentile of wages declines by 0.1 percentage points relative to the pre-reform wage bill, while the 80th percentile of wages declines by 0.2 percentage, and these are the only estimate that are statistically significant.[16] This points to within-firm wage compression as one of the operative mechanisms.

Taken together, these results indicate that the burden of higher property tax liabilities falls disproportionately on higher-earning workers within firms. The reform thus compressed the within-firm wage distribution, reducing inequality between low-and high-wage workers, a pattern that is, in this narrow sense, progressive. These findings relate closely to Risch (2024), who documents that a disproportionate share of the tax burden on business owners is borne by relatively high-earning employees. While property taxes differ from business owner taxes in their statutory incidence, both share the feature of being levied outside of direct business income, suggesting a common transmission mechanism through which firm-level cost shocks are absorbed along the upper tail of the within-firm wage distribution.

VII.A. Heterogeneity Analysis on Firm Survival and Employment

To identify which firms are most vulnerable to increases in property tax liabilities, I estimate heterogeneous treatment effects on firm survival and employment across multiple dimensions. Specifically, I examine heterogeneity by occupancy status (tenant versus owner-occupier), firm size (measured by the number of formal employees), profitability, and leverage. This analysis also sheds light on the mechanisms through which property taxes affect firm outcomes, as well as the distributional consequences of the reform.

Figure 10 summarizes the heterogeneity results on both firm survival and employment. For each dimension, I also estimate event-study coefficients to assess pre-trends within each category. The resulting event-study plots are reported in the Appendix: Figures A.15, A.16, A.17, and A.18 present heterogeneous effects on firm survival, while Figures A.19, A.20, A.21, and A.22 report the corresponding employment effects.

I begin by examining whether the effect on firm survival varies by firms' occupancy status—specifically, whether the firm operates as a tenant or owns land and buildings. This dimension of heterogeneity is motivated by the "first-stage" results in Section V.A,

[16] Appendix Figure A.14 shows a consistent pattern for the 25th, 50th, and 75th percentiles.

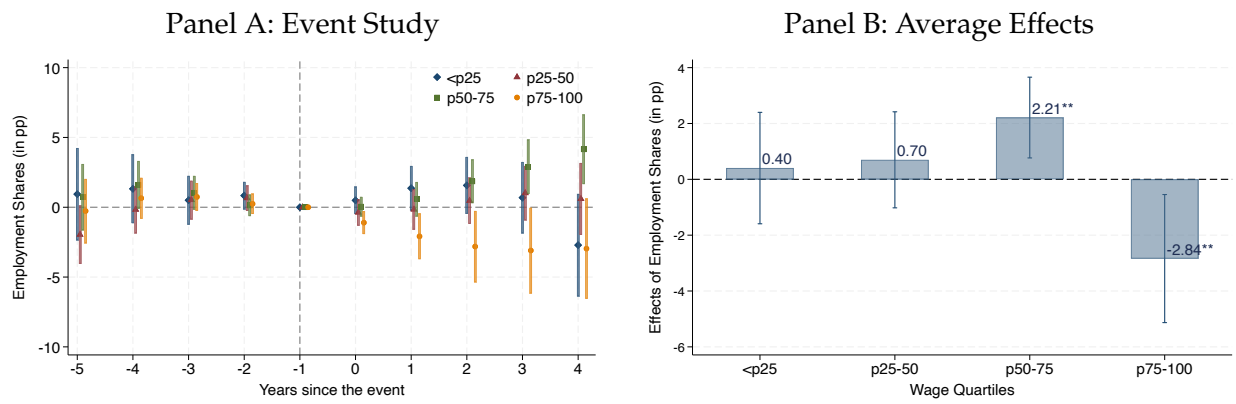


Figure 9: Effects of the Reform on Employment Composition.

Notes: The figure presents the effects of the reform on within firm employment composition. The dependent variables are employment shares in each quartile, relative to pre-reform total employment, i.e., $\frac{E_{i,b,t}}{E_{i,0}} \times 100$, where $E_{i,q,t}$ denotes firm i's number of workers in quartile q at time t and $E_{i,0}$ denotes firm i's total formal employment in the pre-reform period. Accordingly, the coefficient estimates should be interpreted as percentage-point changes relative to the pre-reform total employment. Panel A presents the event-study estimates, with vertical bands indicating the corresponding 95 percent confidence intervals. Panel B presents the average effects. The regression specifications includes fixed effects for sector, pre-reform size, and accounting month. Standard errors are clustered at the firm level.

which show that the reform led to an increase in business rental expenses with substantial pass-through. I classify firms as tenants if they report positive rent expenses in the pre-reform year and as owner-occupiers if they report positive land and building values in their CIT returns. Because some firms report both rent expenses and building values, there is overlap between these categories. For firms reporting neither, I assign occupancy status based on pre-reform capital intensity (the ratio of fixed assets to total assets), classifying firms below the median as tenants and those above the median as owner-occupiers. As a robustness check, I show that excluding firms with imputed occupancy status yields qualitatively similar results, albeit with reduced statistical precision.

Panel A of Figure 10 shows that while the reform reduced formal employment for both tenant and owner-occupier firms, the effect is statistically significant only among tenant businesses. Similarly, Panel B indicates that the negative effects on firm survival probabilities are concentrated among tenant firms. This pattern reinforces the pass-through evidence in Section V.A and further contrasts with the predictions of neoclassical theory. The absence of statistically significant effects on either of these outcomes among firms that report owning land and buildings may reflect scale effects: such firms are likely larger and better able to absorb higher property tax liabilities by adjusting other margins. Consistent with this interpretation, heterogeneity by firm size points in the same direction. Negative

survival effects are concentrated among the smallest firms, those with fewer than five workers. Employment effects are significant only for firms with fewer than five workers and for those employing five to nineteen workers, with no significant effects among larger firms employing twenty or more workers.[17]

To shed light on potential mechanisms, I examine heterogeneity by baseline profitability and leverage. Property taxes raise fixed costs and thereby reduce profitability; firms with thin operating margins may be pushed into losses, increasing their likelihood of exit or employment adjustment. To assess this channel, I first examine effects among loss making firms. I then partition the full sample into terciles and examine how the effects vary across the distribution of pre-reform profitability.[18] Figures 10 shows that the magnitude of both employment and survival effects declines monotonically with profitability. The event-study plots in Appendix Figures A.17 and A.21 show that this pattern is not driven by pre-trends. The finding that survival effects are strongest among loss-making firms further rules out the possibility that the overall survival effects are driven by evasion responses to corporate income tax, since such firms have zero corporate income tax liability.

Turning to leverage, Figure 10 reveals a U-shaped pattern in the employment and survival effects across the leverage distribution: negative effects are statistically significant only among firms at the two tails of the distribution, those with very low and very high leverage, while firms with moderate leverage are largely insulated. This pattern points to two distinct vulnerability channels operating at opposite ends of the distribution. High-leverage firms are exposed through a financial distress channel: higher property tax obligations tighten liquidity against existing debt service commitments, compressing already thin repayment margins and amplifying the probability of distress. At the other end, very low-leverage firms are exposed through a financial exclusion channel: absent meaningful access to credit markets, these firms cannot borrow to smooth the cost shock and are instead forced to adjust on real margins, cutting employment or exiting altogether. Firms with moderate leverage, by contrast, retain sufficient unused debt capacity to absorb the shock through additional borrowing, explaining their relative insulation. This U-shaped incidence pattern suggests that it is financial flexibility, rather than financial strength per se, that shields firms from the employment and survival consequences of higher property tax liabilities.

[17] Although firms employing five to nineteen workers exhibit significant employment effects, the event-study estimates in Appendix Figure A.20 show evidence of pre-trends for this group, complicating a causal interpretation of the average effect. By contrast, there is no evidence of pre-trends for firms with fewer than five workers or for those employing twenty or more workers.

[18] Profit margins are defined as the ratio of profits (EBIT) to revenue.

Appendix Figure A.23 corroborates the above mechanism directly: moderately leveraged firms in treated areas experience a differential increase in debt relative to their counterparts in untreated locations following the reform, consistent with these firms drawing on available credit capacity to smooth the cost shock. No such borrowing response is observed among firms at either tail of the leverage distribution, precisely those for whom the employment and survival effects are most pronounced, reinforcing the interpretation that it is access to the borrowing margin that separates the insulated from the vulnerable.

Taken together, the heterogeneity patterns by profitability and leverage point to financial constraints and financial distress as central mechanisms underlying the reform's effects.

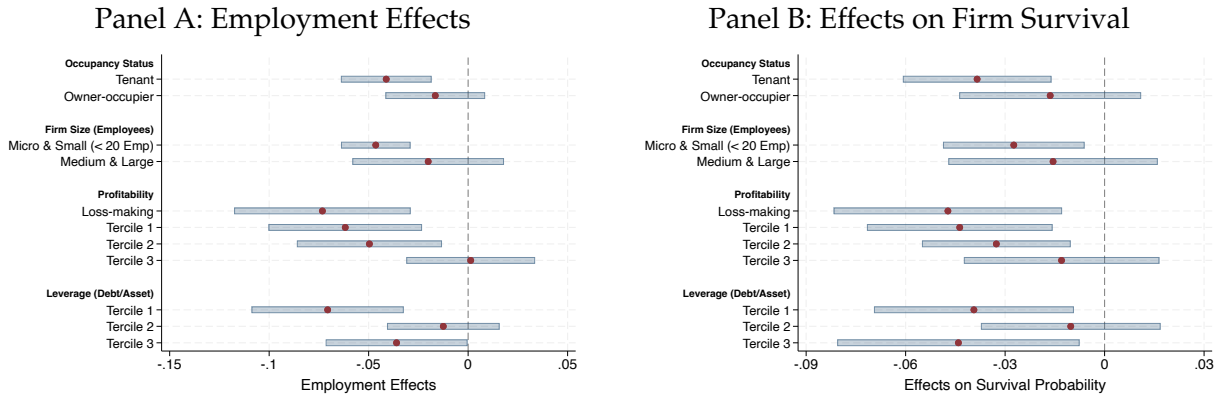


Figure 10: Employment and Survival Effects: Heterogeneity by Firm Characteristics

Notes: The figure shows the heterogeneous effects of the reform on firm survival and employment. Each dot represents a coefficient from a separate regression. Firms are distinguished by pre-reform characteristics based on (i) occupancy status(tenant business or owner-occupier); (ii) firm size (number of formal employees); (iii) profitability (EBIT/revnues); and (iv) leverage (debt-to-asset ratio). The regression specifications includes fixed effects for industry, pre-reform size, and accounting month. Standard errors are clustered at the firm level.

VII.B. Addressing the COVID-19 Threat

A potential threat to my results is that the later years of the sample overlap with the COVID-19 pandemic, which disrupted firm operations and labor markets in ways that could confound the estimated effects of the reform. The identifying assumption required for my estimates to be uncontaminated by COVID-19 is that the pandemic did not differentially affect treated and comparison firms. This assumption is plausible for two reasons. First, I restrict never-treated comparison units to firms located just outside the city boundary, which are likely to have faced similar pandemic exposure as treated firms within the

city. Second, in the main analysis I use not-yet-treated firms, which includes both never-treated firms and those located within the city but treated at a later date, as the comparison group, further limiting the scope for differential COVID-19 exposure across groups. Reassuringly, the estimates are comparable in magnitude and precision when using only never-treated firms — those located outside the city boundary — as the comparison group instead.

Nevertheless, I conduct two additional checks to rule out the possibility that COVID-19 drives the results. First, I restrict the sample to the pre-COVID period by dropping all observations after 2018/19. Panels A and C of Figure A.24 show that differential declines in employment and wages persist in this shortened panel. Although the average effects are smaller in magnitude, the event-study plots reveal that the effects were still accumulating by the end of the truncated sample, suggesting that the full-sample estimates reflect the continuation of a trend that predates the pandemic, rather than a COVID-induced shock, and confirming that the results are not an artifact of the pandemic period.

Second, I exploit the calendar-time specific effects estimator embedded in Callaway and Sant'Anna (2021)'s staggered difference-in-differences framework, which recovers the average treatment effect for each calendar year separately, using only firms treated as of that date. This allows me to trace whether the employment and wage effects emerge prior to the COVID-19 period or materialize only during the pandemic. Panel B of Figure A.24 shows that employment effects were already building before the onset of the pandemic, with estimates growing in magnitude through 2018/19 and persisting into the COVID period. Panel D shows an identical pattern for wages. The fact that the effects predate the pandemic makes it implausible that COVID-19 is the primary driver of the estimated impacts.

VIII. IMPLICATIONS FOR CENTRAL GOVERNMENT TAX REVENUE

The firm-level effects documented in Sections VI.A, VI.B, and VI.C have direct implications for central government revenues. To the extent that the reform reduces firm profitability, induces firm exit, and contracts formal employment and wages, it erodes the tax bases of both the corporate income tax and the employee income tax (Pay-As-You-Earn, PAYE), potentially offsetting a share of the property tax revenues raised by the local government. This fiscal externality represents a form of vertical tax competition: when a sub-national government raises property taxes, it may inadvertently compress the tax bases of taxes administered at the national level, generating a negative fiscal spillover that is invisible to the local authority setting the rate but consequential for consolidated

public revenues.[19] Whether and to what extent such spillovers materialize is ultimately an empirical question. In this section, I examine how central government tax revenues evolve around the reform to speak directly to this question.

Figure 11 presents the event-study estimates and average effects on central government tax revenues, measured in millions of Uganda shillings. I examine three outcomes in turn: corporate income tax payments (Panel A), employee income tax (PAYE) payments aggregated to the firm level (Panel B), and the combined revenue from both taxes (Panel C).[20] Despite inducing firm exit, and profit declines among a subset of firms, the reform had no statistically significant effect on corporate income tax revenues. This result is consistent with the earlier finding that the firms most likely to exit were precisely those with minimal or no tax liability — loss-making and least profitable firms that contributed little to the corporate tax base even before the reform.

The picture is different for employee income tax revenues. Panel B reveals a differential decline in PAYE receipts from treated relative to untreated firms following the reform, driven by the contraction in formal employment and wages and the increase in wage informality documented in Section VI.C. These labor market adjustments translate directly into a smaller formal wage bill from which PAYE is withheld, reducing central government receipts.

Combining both taxes, Panel C shows that total central government revenue collected from treated firms declines by an average of UGX 0.5 million annually relative to untreated firms, against a pre-reform mean of UGX 7.8 million — a decline of 6.4 percent. As reported in Appendix Table A.5, this is equivalent to UGX 0.25 of lost central government revenue for every UGX 1 increase in property tax revenues, suggesting that the fiscal gains at the local level are substantially offset by revenue losses at the central level.

To assess how much of the decline in central government PAYE revenues is attributable to wage informalization rather than outright job destruction, I combine the estimated increase in informal wage shares with pre-reform wage bill levels and the average effective PAYE rate to compute the implied PAYE loss from informalization alone. This back-of-the-envelope calculation suggests that the increase in wage informality documented in Section VI.C accounts for approximately 43 percent of the estimated decline in PAYE revenues.[21] The remaining share reflects the mechanical reduction in the formal wage bill from employment losses.

[19] This vertical fiscal externality is distinct from the horizontal tax competition literature, which focuses on competition between jurisdictions at the same level of government (see, e.g., Wilson 1999).

[20] I aggregate employee taxes at firm level because the data does not allow me to follow workers overtime, as workers are only uniquely identified at firm level.

[21] Appendix III.A shows the detailed steps of the back-of-the-envelope calculation.

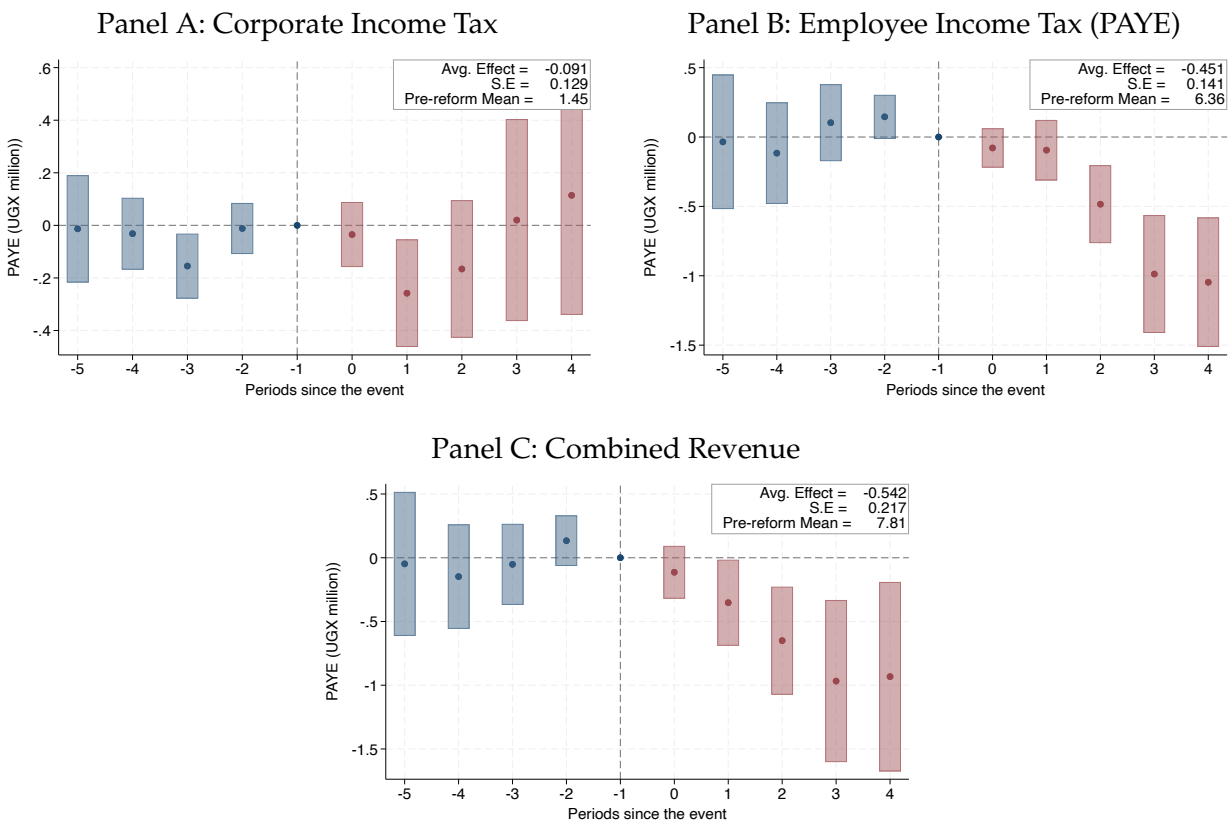


Figure 11: Implications for Central Government Tax Revenue

Notes: The figure presents the dynamic effects of the reform on central government tax revenue. All the analysis is aggregated at firm level. The outcome variable is in millions of Uganda shillings (UGX), constant 2016/17. Panel A shows the effects on corporate income tax, Panel B shows the effects on employee tax revenue (PAYE), while Panel C shows the effects on the combined revenue. Each dot denotes the point estimate of the treatment effect for that year, with the reference period set at -1. The vertical bars indicate the corresponding 90 percent confidence intervals. The regression specification includes fixed effects for pre-reform industry-size and accounting month. Standard errors are clustered at the firm level.

IX. CONCLUSION

This paper examines the effects of commercial property taxes on real activity and the distributional consequences at firm level. I exploit a natural experiment in Kampala, the capital city of Uganda, where property tax liabilities increased following a revaluation and mass property enumeration exercise, while surrounding areas were unaffected. This setting allows me to estimate causal effects using a difference-in-differences identification strategy. I use data from corporate income tax returns and employee tax returns.

First, I show that the reform increased business rental costs by 9 percent, with an estimated citywide pass-through rate of 0.42, sharply contradicting the neoclassical prediction that landlords should bear the entire burden. To explain this contrast, I propose a

framework that embeds a key institutional feature of selective enforcement: the probability of enforcement, and therefore compliance, is higher when a property is occupied and generating observable economic activity, and lower when the unit is unoccupied. Second, I show that while the reform had no significant effects on revenues, it reduced firms' profitability, investment, employment and survival probabilities.

Finally, I examine the distributional consequences of the reform by assessing which types of workers are most affected. First, I study how within-firm wage quantiles evolve before and after the reform in treated firms relative to untreated firms. Second, I examine how the share of workers, relative to pre-reform total employment, in fixed wage quartiles evolves following the reform. The evidence indicates that workers at the top of the earnings distribution are disproportionately affected. In particular, following the reform, the 75th percentile of wages declines differentially in treated firms relative to untreated firms, while there is no statistically significant effect on the 25th and 50th percentiles, despite negative point estimates. Consistent with this pattern, I also find a decline in the share of workers in the top quartile of the pre-reform wage distribution, alongside an increase in the share of workers in the 50th–75th percentile. Taken together, these results suggest that the reform reduced within-firm wage inequality and was therefore progressive, a pattern consistent with skill downgrading within firms.

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I. APPENDIX: ADDITIONAL FIGURES AND TABLES

I.A. Additional Figures

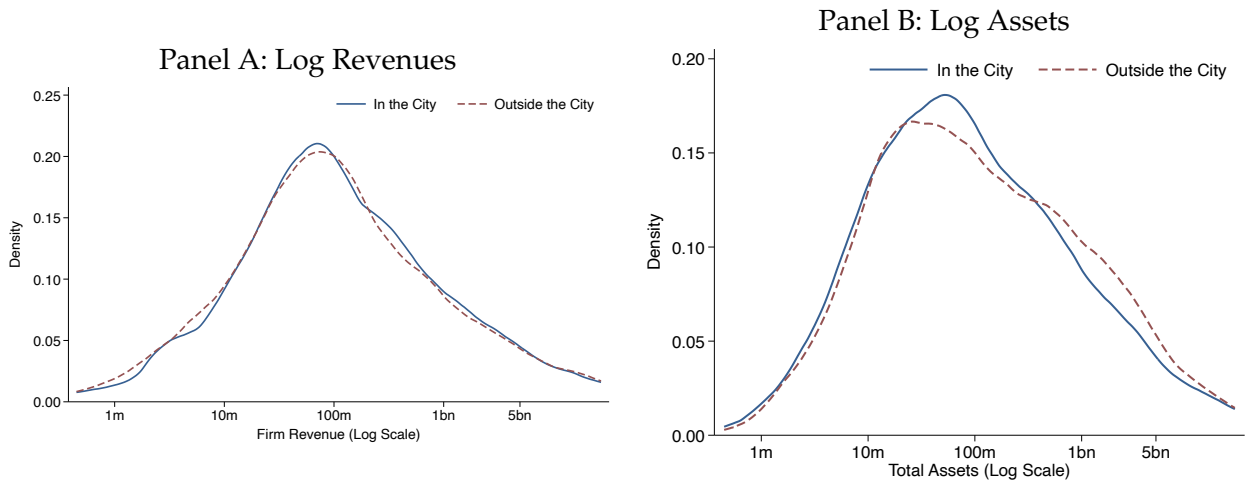

Figure A.1: Firm Size Distributions Across Locations

Notes: This figure plots the size distribution (in log scale) of pre-reform active firms located inside (treated) and outside (comparison) the city. Panels A shows the distribution of revenues while Panel B shows the distribution of assets.

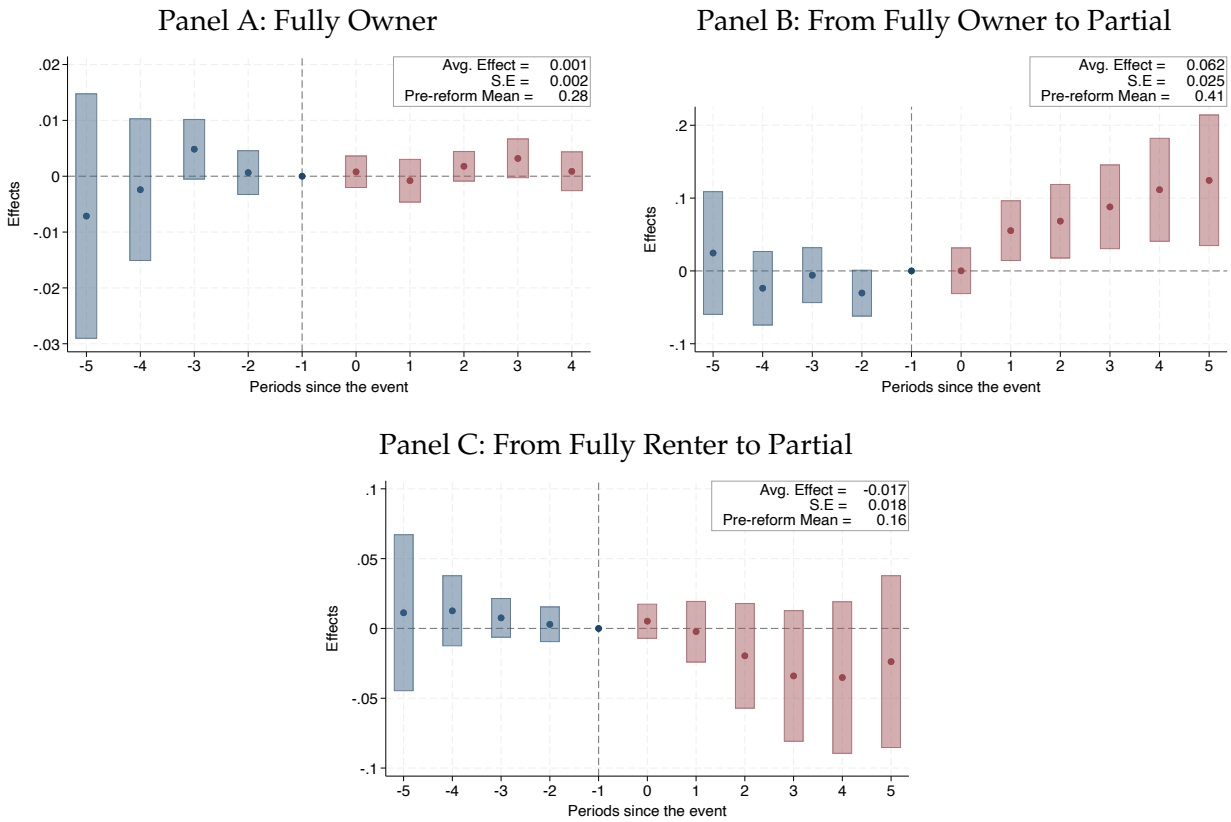

Figure A.2: Effects of the Reform on Occupancy Decision: *Own or Rent?*

Notes: The figure shows the extensive margin response of firms' occupancy decisions following the reform. Panel A examines the binary choice between full ownership and full tenancy: the outcome takes the value one if the firm reports positive land and building values with zero rent expenses (fully owned), and zero if the firm reports positive rent expenses with zero land and building values (fully rented). This panel excludes firms reporting positive values in both categories. Panel B examines the transition from full ownership toward partial occupancy: the outcome takes the value one if the firm reports positive values in both land and buildings and rent expenses, and zero if the firm reports positive values in land and buildings only. Panel C examines the transition from full tenancy toward partial occupancy: the outcome takes the value one if the firm reports positive values in both rent expenses and land and buildings, and zero if the firm reports positive values in rent expenses only. Together, Panels B and C identify whether the reform induced firms to move toward partial occupancy arrangements — simultaneously owning and leasing space — from a previously full ownership or full tenancy position. The horizontal axis shows the number of years since the policy. Each dot represents the point estimate of the treatment effect for that year, with the reference category set at -1. The vertical bands indicate the corresponding 90 percent confidence intervals. The coefficients are estimated using a staggered Difference-in-Differences approach, following Callaway and Sant'Anna (2021), with not-yet treated units used as the comparison group. Standard errors are clustered at the firm level.

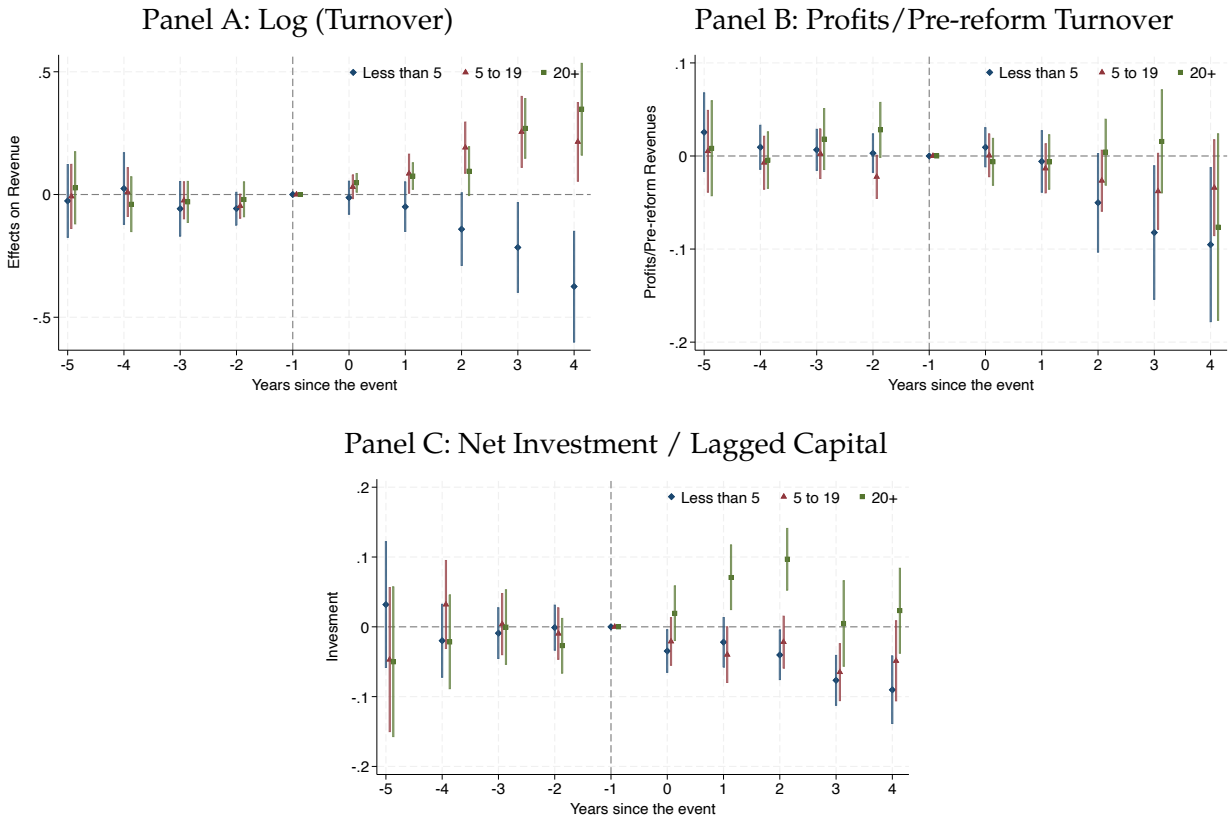

Figure A.3: Effects on Revenues, Profits, and Investment: *Heterogeneity by Size*

Notes: The Figure presents the effects of the reform on revenues, profits, and investment for different firm sizes (measured by the number of formal workers. Revenues are transformed in logs. Profits are defined as earnings before interest and taxes (EBIT), computed as sales less cost of sales, operating and administration expenses. The calculated profits are then scaled by pre-reform revenues. Investment rate is defined as the annual change in net fixed assets, scaled by lagged fixed assets. The horizontal axis indicates the number of years since the reform. Each dot represents the point estimate of the treatment effect for that year, with the reference category set at -1. The vertical bands indicate the corresponding 90 percent confidence intervals. The regression specification includes fixed effects for sector and pre-reform size, where applicable. The coefficients are estimated using a staggered Difference-in-Differences approach, following Callaway and Sant'Anna (2021), with not-yet treated units used as the comparison group. Standard errors are clustered at the firm level.

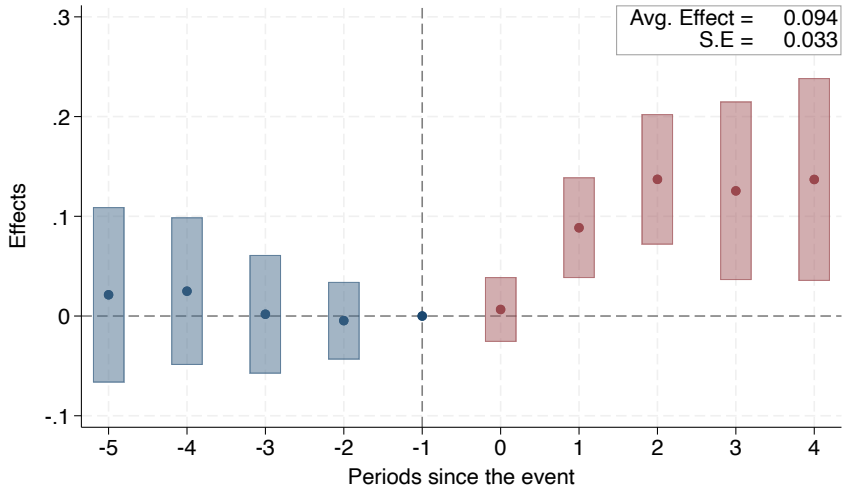

Figure A.4: Effects of Property Taxes on Business Rental Expenses: Never Treated as Comparison Group

Notes: The figure shows the dynamic effects of property taxes on business rental expenses. The dependent variable is rent costs scaled by its pre-reform value, implying the coefficients can be interpreted as a percent change after multiplying by 100. The horizontal axis shows the number of years since the policy. Each dot represents the point estimate of the treatment effect for that year, with the reference category set at -1. The vertical bands indicate the corresponding 90 percent confidence intervals. The regression specification includes fixed effects for sector and 2016 size bins. The coefficients are estimated using a staggered Difference-in-Differences approach, following Callaway and Sant'Anna (2021), with never treated units used as the comparison group. Standard errors are clustered at the firm level.

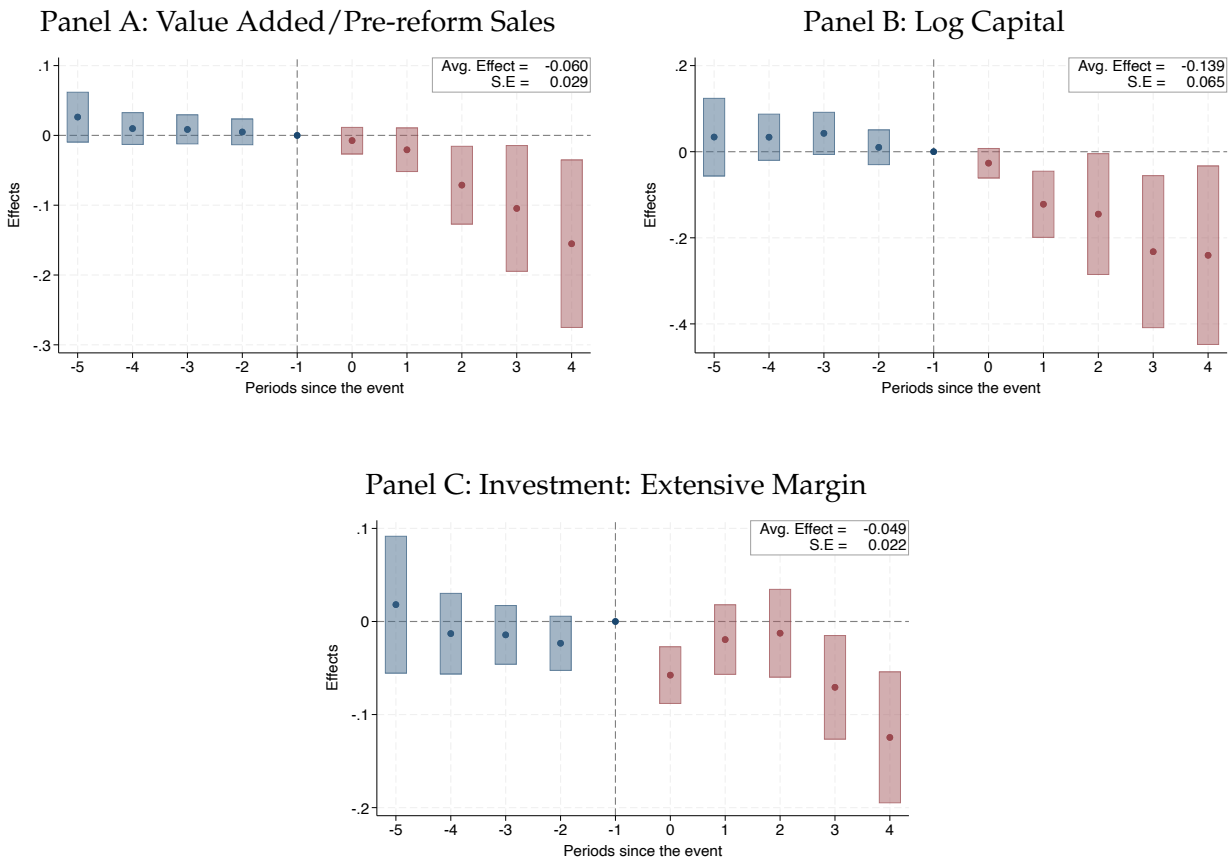

Figure A.5: Effects on Value Added, Capital, and Extensive Margin Investment

Notes: The Figure presents the dynamic effects of the reform on firms' value added, capital (in logs) and investment (extensive margin). Value added is defined as revenues minus cost of sales, normalized by pre-reform sales. Capital is defined as the book value of net fixed assets. Investment is defined as the annual change in fixed assets. At the extensive margin, the outcome is one if the firm had positive investment, zero otherwise. The horizontal axis indicates the number of years since the reform. Each dot represents the point estimate of the treatment effect for that year, with the reference category set at -1. The vertical bands indicate the corresponding 90 percent confidence intervals. The regression specification includes fixed effects for sector and 2016 size bins. The coefficients are estimated using a staggered Difference-in-Differences approach, following Callaway and Sant'Anna (2021), with not-yet treated units used as the comparison group. Standard errors are clustered at the firm level.

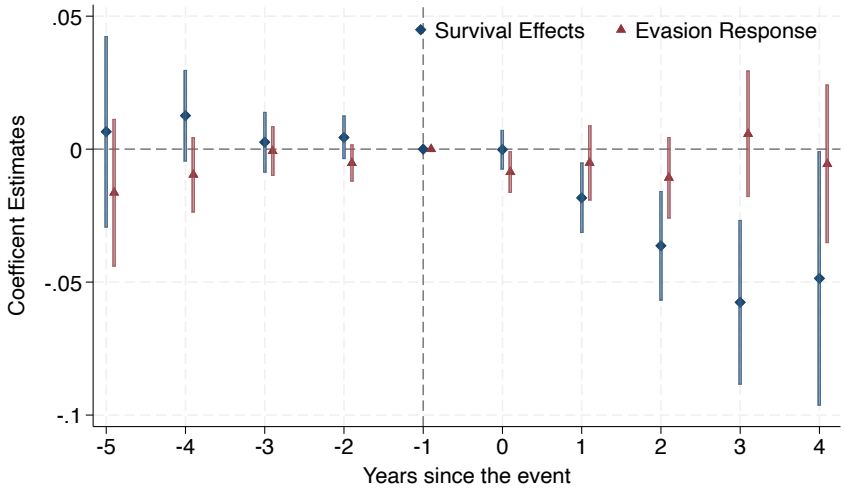

Figure A.6: Effects on Firm Survival vs Evasion

Notes: The figure presents the dynamic effects of the reform on firm survival, separating survival from evasion response. The outcome variable is an indicator equal to one if the firm is operating at time t, zero otherwise, implying the coefficients can be interpreted as percentage point change in survival probability. The horizontal axis indicates the number of years since the reform. Each dot represents the point estimate of the treatment effect for that year, with the reference category set at -1. The vertical bands indicate the corresponding 90 percent confidence intervals. The regression specification includes fixed effects for sector and 2016 size bins. The coefficients are estimated using a staggered Difference-in-Differences approach, following Callaway and Sant'Anna (2021), with not-yet treated units used as the comparison group. Standard errors are clustered at the firm level.

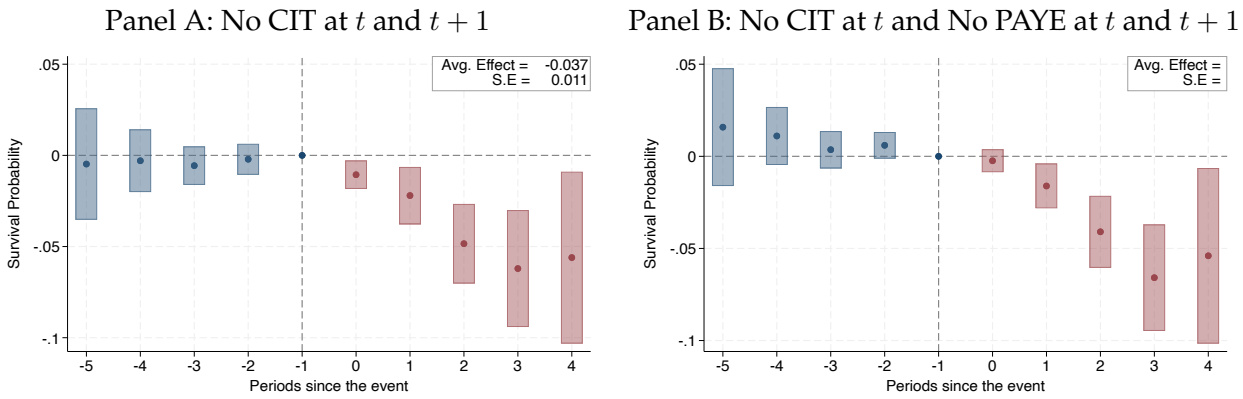

Figure A.7: Property Tax Enforcement and Firm Survival: Robustness at t and $t+1$

Notes: The figure presents the dynamic effects of the reform on firm survival. In Panel A I define exit as the absence of corporate income tax (CIT) return for at least two consecutive year. In Panel B restricts exit to firms that have no CIT return in year t and no footprints in the employee tax returns in both year t and $t+1$. The horizontal axis indicates the number of years since the reform. Each dot represents the point estimate of the treatment effect for that year, with the reference category set at -1. The vertical bands indicate the corresponding 90 percent confidence intervals. The regression specification includes fixed effects for sector, pre-reform size, and accounting month. The coefficients are estimated using a staggered Difference-in-Differences approach, following Callaway and Sant'Anna (2021), with not-yet treated units used as the comparison group. Standard errors are clustered at the firm level.

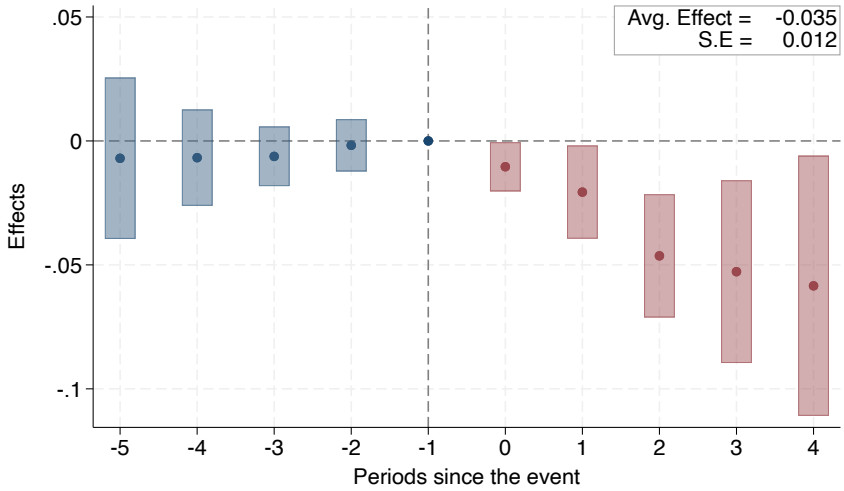

Figure A.8: Property Tax Enforcement and Firm Survival: Sample of Firms Present in First Year

Notes: The figure presents the dynamic effects of the reform on firm survival, with the sample fixed to firms active in the first year of the sample rather than the year immediately prior to the reform. The outcome variable is an indicator equal to one if the firm is operating at time t, zero otherwise, implying the coefficients can be interpreted as percentage point change in survival probability. The horizontal axis indicates the number of years since the reform. Each dot represents the point estimate of the treatment effect for that year, with the reference category set at -1. The vertical bands indicate the corresponding 90 percent confidence intervals. The regression specification includes fixed effects for sector, pre-reform size, and accounting month. The coefficients are estimated using a staggered Difference-in-Differences approach, following Callaway and Sant'Anna (2021), with not-yet treated units used as the comparison group. Standard errors are clustered at the firm level.

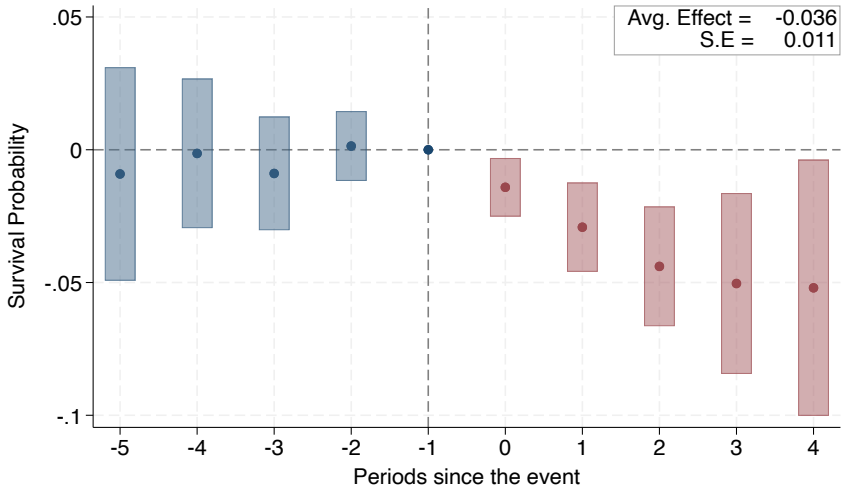

Figure A.9: Effects of Property Tax Enforcement on Firm Survival: Never-Treated as Comparison Group

Notes: The figure presents the dynamic effects of the reform on firm survival, using never treated units as the comparison group rather than not-yet treated. The outcome variable is an indicator equal to one if the firm is operating at time t, zero otherwise, implying the coefficients can be interpreted as percentage point change in survival probability. The horizontal axis indicates the number of years since the reform, with negative values representing pre-treatment periods. Each dot denotes the point estimate of the treatment effect for that year, with the reference period set at -1. The vertical bands indicate the corresponding 90 percent confidence intervals. The regression specification includes fixed effects for sector, pre-reform size, and accounting month. The coefficients are estimated using a staggered Difference-in-Differences approach following Callaway and Sant'Anna (2021), using never-treated units as the comparison group. Standard errors are clustered at the firm level.

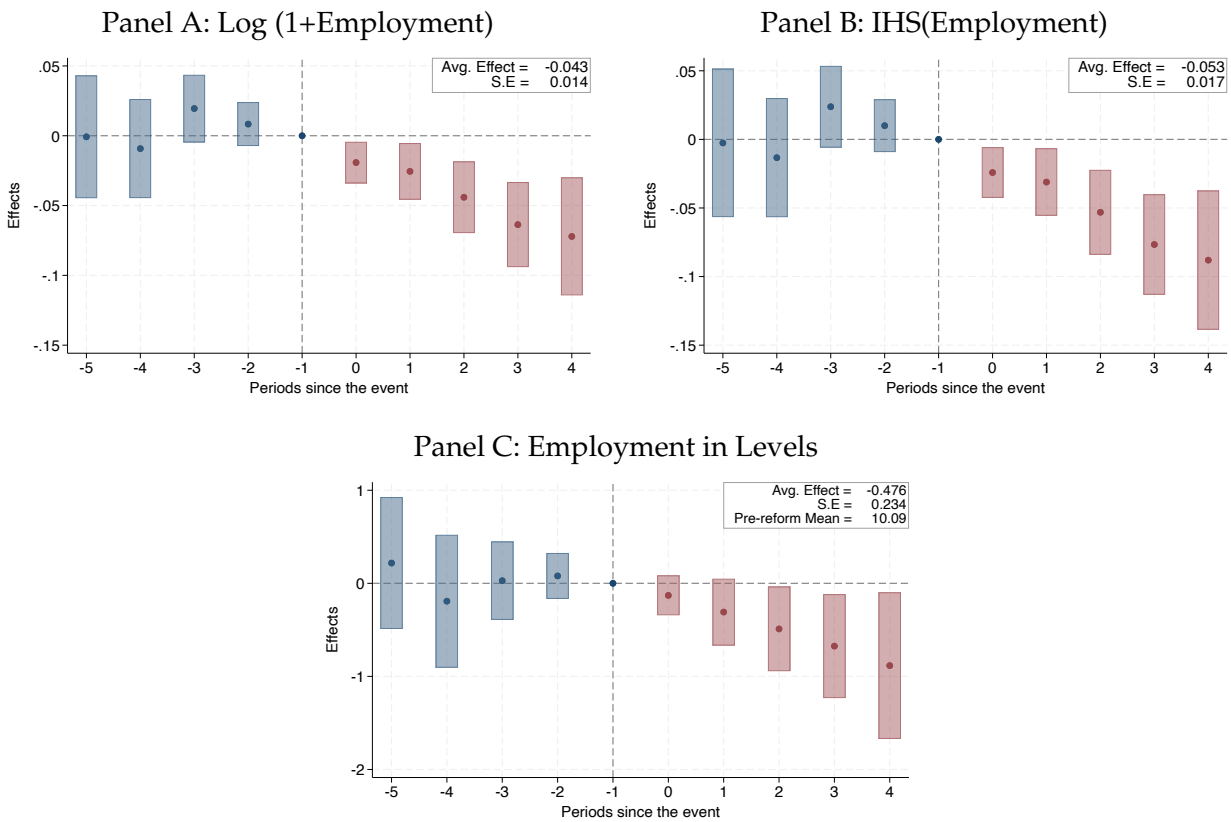

Figure A.10: Effects on Formal Employment: Logs, IHS, and Levels

Notes: The figure presents the dynamic effects of the reform on firms' formal employment. In Panel A employment is transformed using logs, i.e log(1+employment). In Panel B employment is transformed using inverse hyperbolic sine transformation. Panel C presents the effects in levels where the scale on the vertical axis is number of jobs. Each dot denotes the point estimate of the treatment effect for that year, with the reference period set at -1. The vertical bars indicate the corresponding 90 percent confidence intervals. The regression specification includes fixed effects for pre-reform industry-size and accounting month. Standard errors are clustered at the firm level.

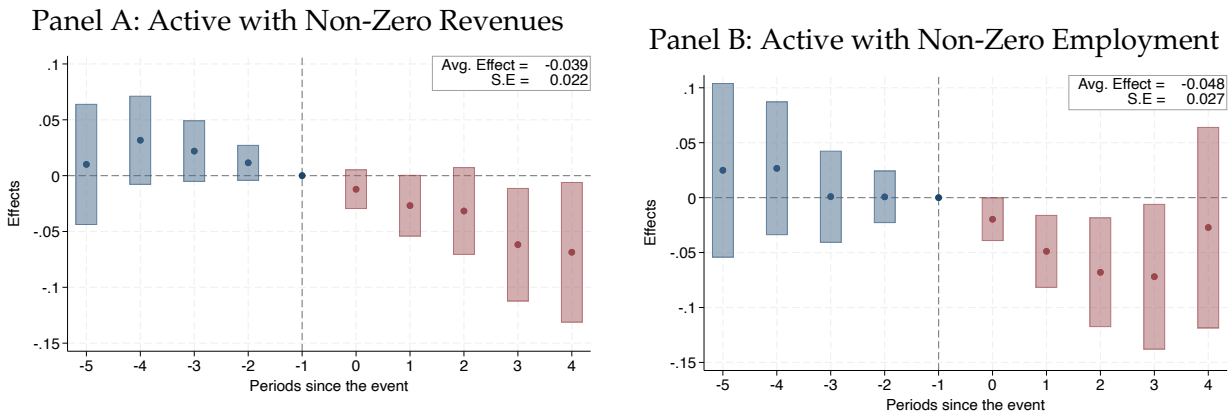

Figure A.11: Effects on Formal Employment: Sample of Firms Active in all Sample Years

Notes: The figure presents the dynamic effects of the reform on formal employment with the sample restricted to a panel of firms that are active in all the years of the sample. Panel A restricts the sample to firms with non-zero revenues in all years, normalizing the dependent variable by its pre-reform value. Panel B restricts the sample to firms with non-zero workers in all years, transforming the outcome variable using logs. Each dot denotes the point estimate of the treatment effect for that year, with the reference period set at -1. The vertical bars indicate the corresponding 90 percent confidence intervals. The regression specification includes fixed effects for sector, pre-reform size, and accounting month. The coefficients are estimated using a staggered Difference-in-Differences approach following Callaway and Sant'Anna (2021). Standard errors are clustered at the firm level.

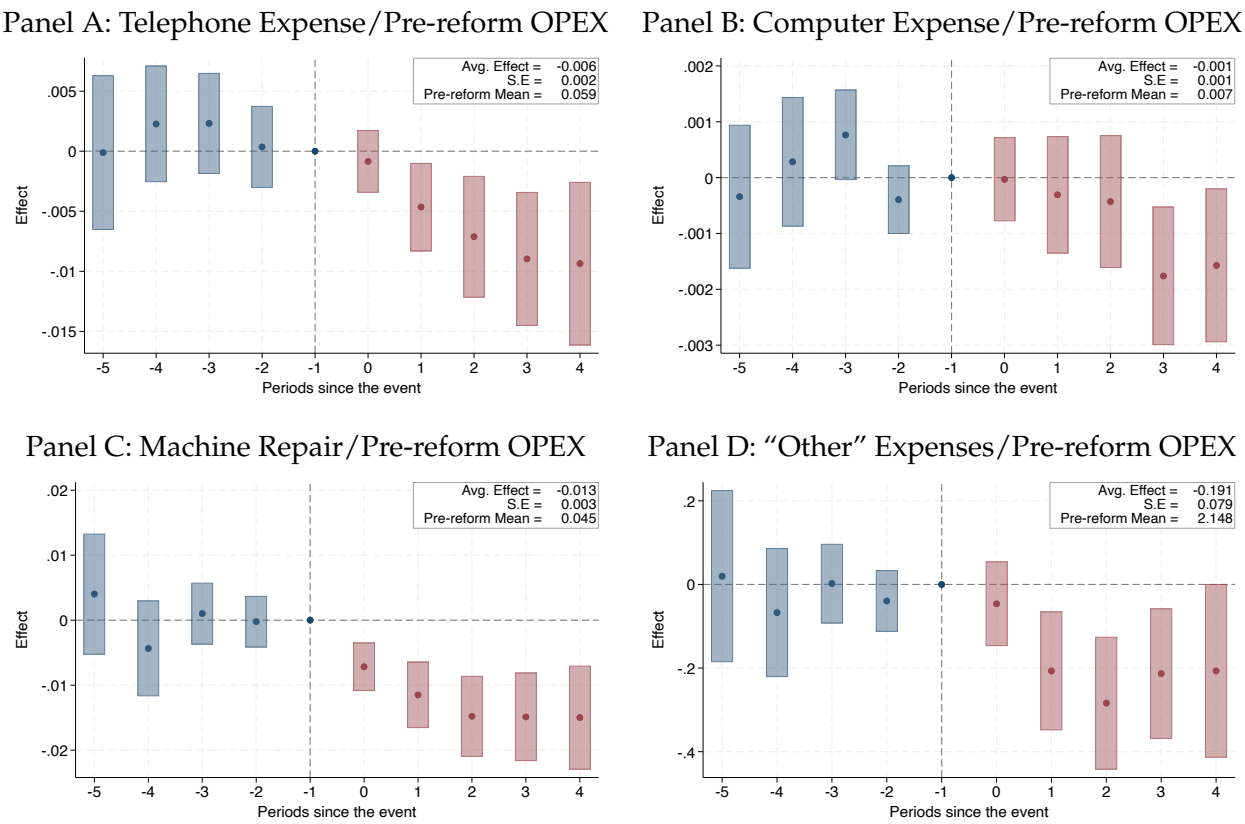

Figure A.12: Effects on Different Categories of Operating Expenses

Notes: The figure presents the dynamic effects of the reform on firms' formal employment. In Panel A employment is transformed using logs, i.e log(1+employment). In Panel B employment is transformed using inverse hyperbolic sine transformation. Panel C presents the effects in levels where the scale on the vertical axis is number of jobs. Each dot denotes the point estimate of the treatment effect for that year, with the reference period set at -1. The vertical bars indicate the corresponding 90 percent confidence intervals. The regression specification includes fixed effects for pre-reform industry-size and accounting month. Standard errors are clustered at the firm level.

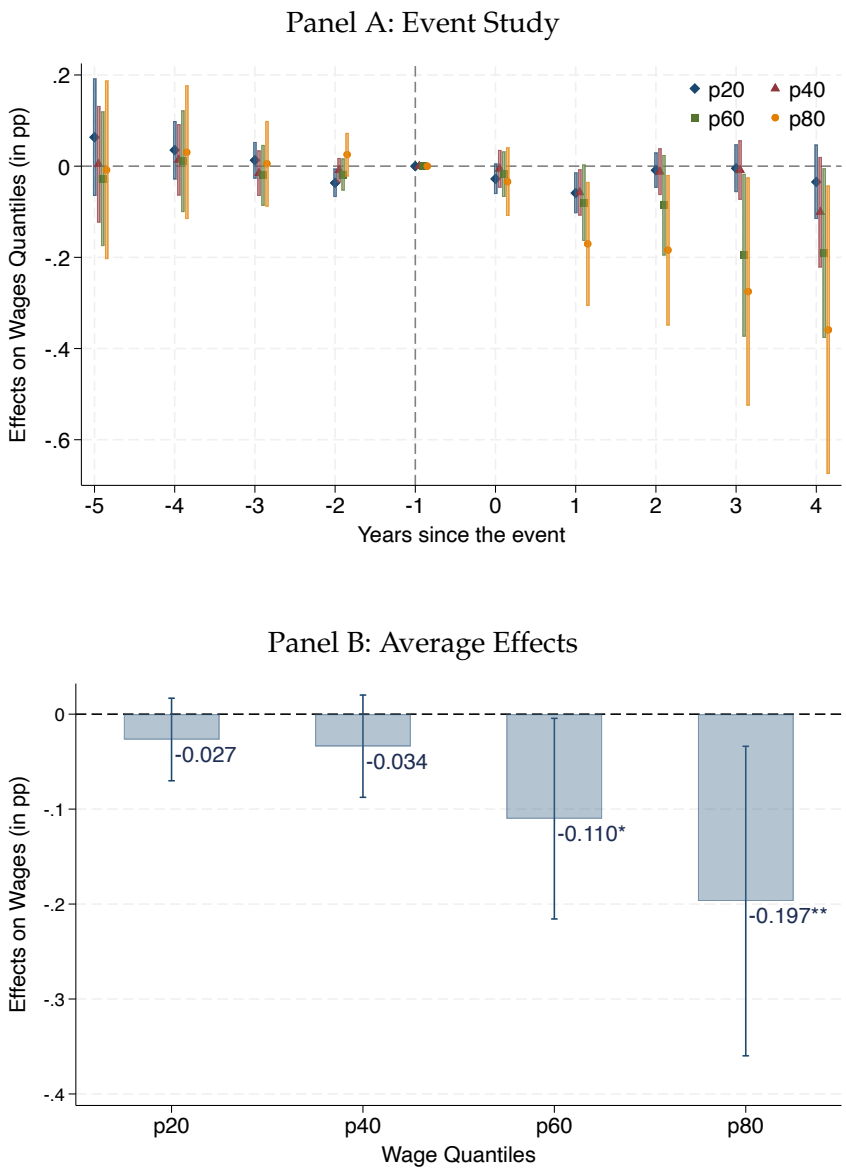

Figure A.13: Effects on Wage Quantiles: 20th, 40th, 60th, and 80th percentiles

Notes: The figure presents the effects of the reform on within-firm wage quantiles. The dependent variables are the 20th, 40th, 60th, and 80th within-firm percentiles of wages, scaled by the pre-reform formal wage bill, i.e., $\frac{q_{i,t}}{W_{i,0}} \times 100$, where $q_{i,t}$ denotes firm i's q^{th} wage quantile at time t and $W_{i,0}$ denotes firm i's pre-reform wage bill. Accordingly, the coefficient estimates should be interpreted as percentage-point changes relative to the pre-reform wage bill. Panel A presents the event-study estimates, with vertical bands indicating the corresponding 95 percent confidence intervals. Panel B presents the average effects. The regression specification includes fixed effects for sector, pre-reform size, and accounting month. The coefficients are estimated using a staggered difference-in-differences design following Callaway and Sant'Anna (2021). Standard errors are clustered at the firm level.

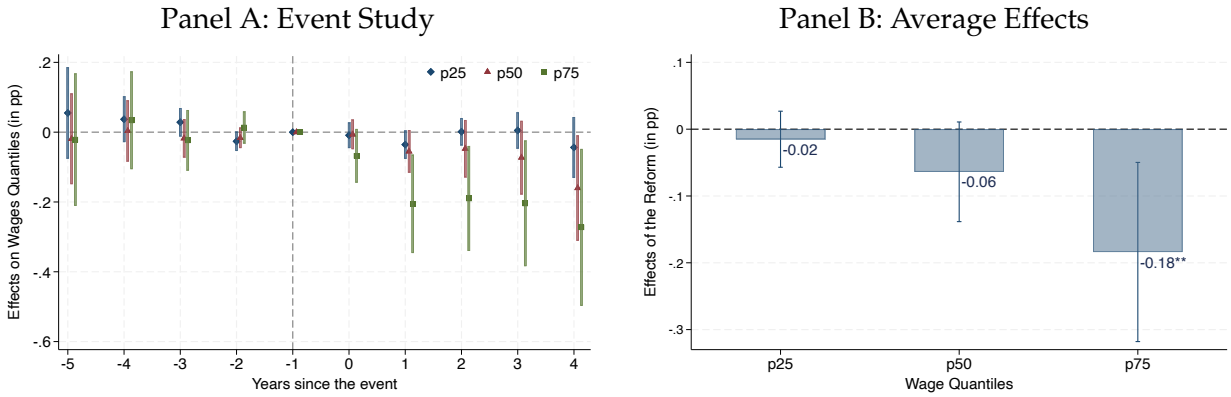

Figure A.14: Effects on Wage Quantiles: Percentage Points Relative to Pre-reform Wage Bill

Notes: The figure presents the effects of the reform on within-firm wage quantiles. The dependent variables are the 25th, 50th, and 75th within-firm percentiles of wages, scaled by the pre-reform formal wage bill, i.e., $\frac{q_{i,t}}{W_{i,0}} \times 100$, where $q_{i,t}$ denotes firm i's q^{th} wage quantile at time t and $W_{i,0}$ denotes firm i's pre-reform wage bill. Accordingly, the coefficient estimates should be interpreted as percentage-point changes relative to the pre-reform wage bill. Panel A presents the event-study estimates, with vertical bands indicating the corresponding 95 percent confidence intervals. Panel B presents the average effects. The regression specifications includes fixed effects for sector, pre-reform size, and accounting month. Standard errors are clustered at the firm level.

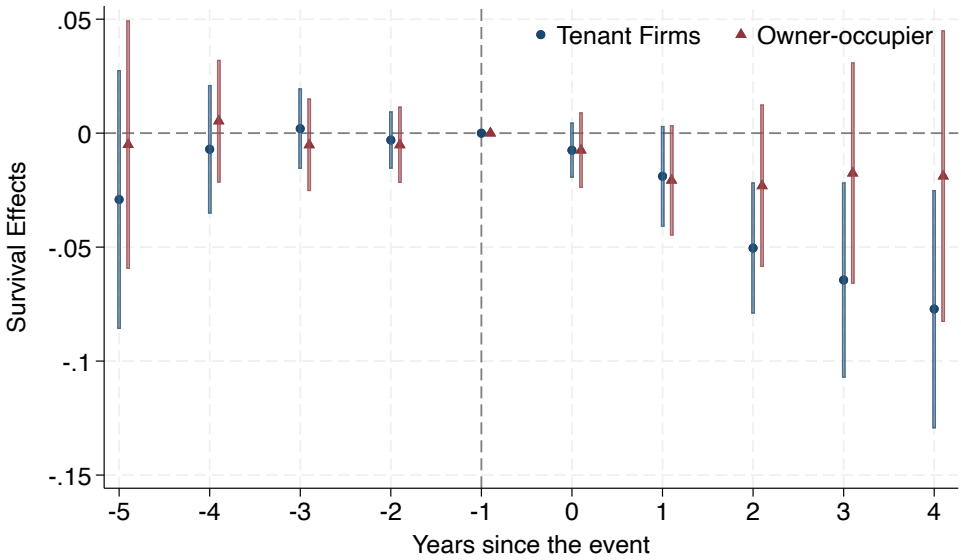

Figure A.15: Survival Effects: Heterogeneity by Occupancy Status

Notes: The figure shows the dynamic effects of the reform on firm survival, separately for tenant and owner-occupier firms. Firms are classified as tenants if they report positive rent expenses in the pre-reform year and as owner-occupiers if they report positive land and building values in their CIT returns. Firms reporting neither are assigned ownership status based on pre-reform capital intensity, with those below (above) the median classified as tenants (owner-occupiers). The outcome variable is an indicator equal to one if the firm is operating at time t, zero otherwise, implying the coefficients can be interpreted as percentage point change in survival probability. The horizontal axis indicates the number of years since the reform. Each dot represents the point estimate of the treatment effect for that year, with the reference category set at -1. The vertical bands indicate the corresponding 95 percent confidence intervals. The regression specification includes fixed effects for sector, pre-reform size, and accounting month. The coefficients are estimated using a staggered Difference-in-Differences approach, following Callaway and Sant'Anna (2021), with not-yet treated units used as the comparison group. Standard errors are clustered at the firm level.

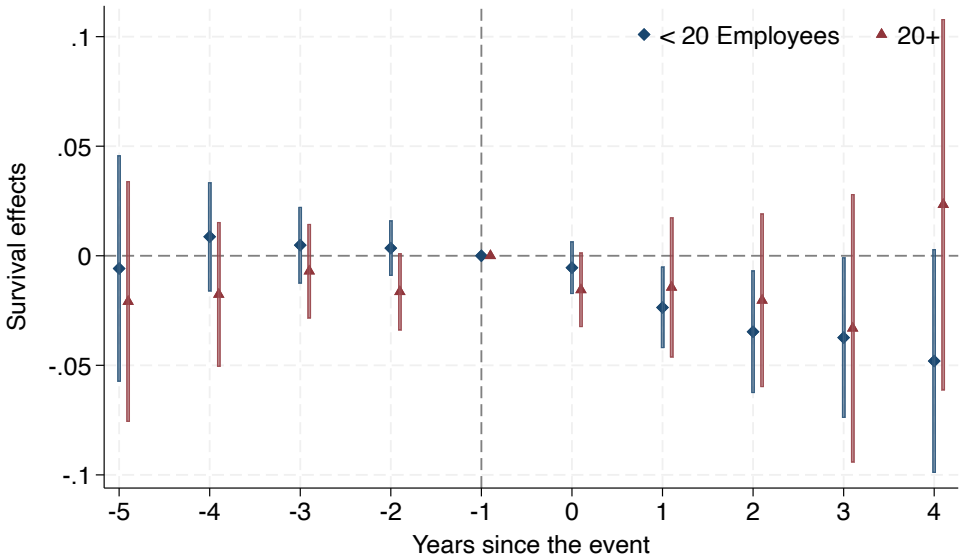

Figure A.16: Survival Effects: Heterogeneity by Size (Number of Formal Employees)

Notes: The figure shows the dynamic effects of the reform on firm survival, separately for different firm sizes based on their pre-reform formal employment. The outcome variable is an indicator equal to one if the firm is operating at time t, zero otherwise, implying the coefficients can be interpreted as percentage point change in survival probability. The horizontal axis indicates the number of years since the reform. Each dot represents the point estimate of the treatment effect for that year, with the reference category set at -1. The vertical bands indicate the corresponding 95 percent confidence intervals. The regression specification includes fixed effects for sector, pre-reform size, and accounting month. The coefficients are estimated using a staggered Difference-in-Differences approach, following Callaway and Sant'Anna (2021), with not-yet treated units used as the comparison group. Standard errors are clustered at the firm level.

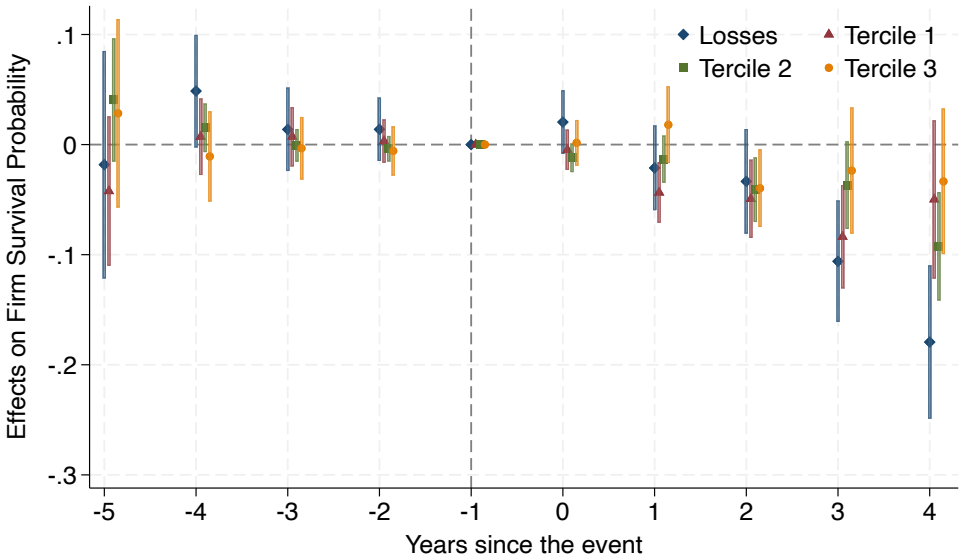

Figure A.17: Survival Effects: Heterogeneity by Profitability

Notes: The figure shows the dynamic effects of the reform on firm survival for firms at different points in the pre-reform profit margins (EBIT/revenues) distribution. The outcome variable is an indicator equal to one if the firm is operating at time t, zero otherwise, implying the coefficients can be interpreted as percentage point change in survival probability. The horizontal axis indicates the number of years since the reform. Each dot represents the point estimate of the treatment effect for that year, with the reference category set at -1. The vertical bands indicate the corresponding 95 percent confidence intervals. The regression specification includes fixed effects for sector, pre-reform size, and accounting month. The coefficients are estimated using a staggered Difference-in-Differences approach, following Callaway and Sant'Anna (2021), with not-yet treated units used as the comparison group. Standard errors are clustered at the firm level.

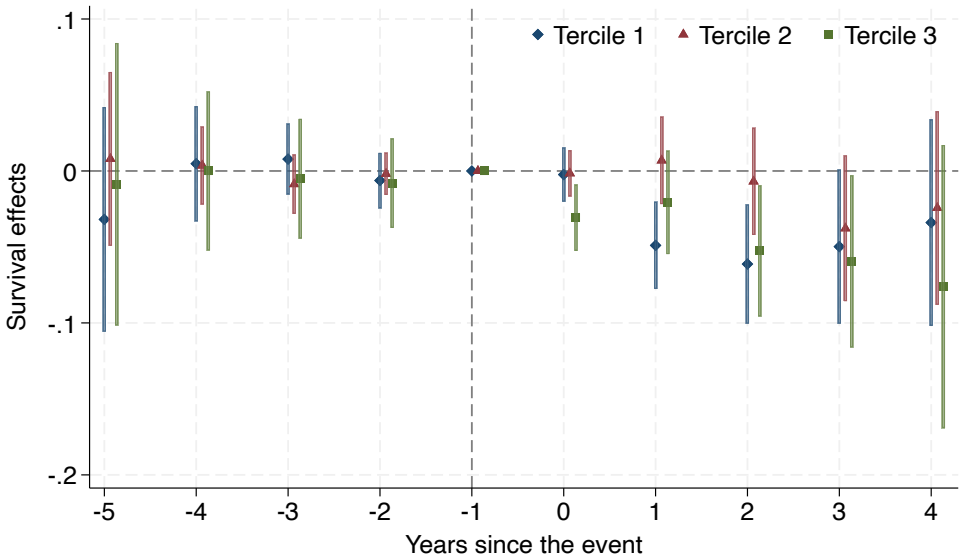

Figure A.18: Survival Effects: Heterogeneity by Level of Leverage

Notes: The figure shows the dynamic effects of the reform on firm survival for firms with different levels of pre-reform leverage (debt-to-asset ratio). The outcome variable is an indicator equal to one if the firm is operating at time t, zero otherwise, implying the coefficients can be interpreted as percentage point change in survival probability. The horizontal axis indicates the number of years since the reform. Each dot represents the point estimate of the treatment effect for that year, with the reference category set at -1. The vertical bands indicate the corresponding 95 percent confidence intervals. The regression specification includes fixed effects for sector, pre-reform size, and accounting month. The coefficients are estimated using a staggered Difference-in-Differences approach, following Callaway and Sant'Anna (2021), with not-yet treated units used as the comparison group. Standard errors are clustered at the firm level.

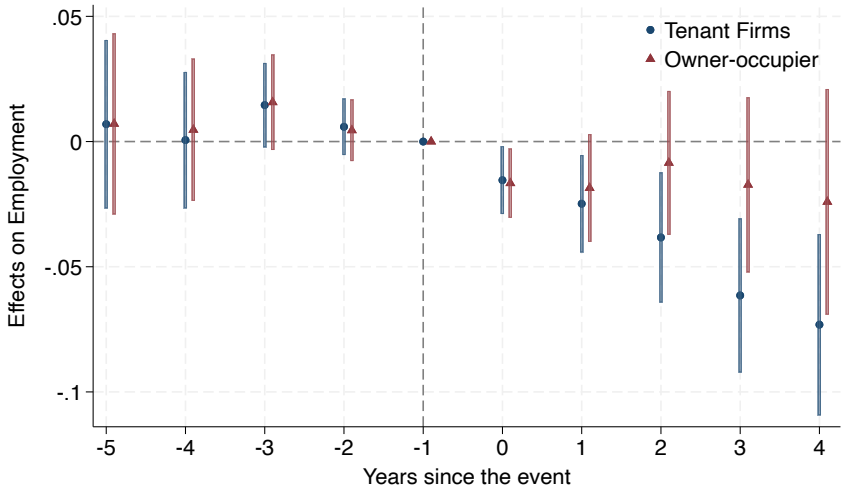

Figure A.19: Employment Effects: Heterogeneity by Occupancy Status

Notes: The figure shows the dynamic effects of the reform on formal employment, separately for tenant and owner-occupier firms. The outcome variable is number of workers, normalized by its pre-reform value, implying the coefficients can be interpreted as a percent changes. Firms are classified as tenants if they report positive rent expenses in the pre-reform year and as owner-occupiers if they report positive land and building values in their CIT returns. Firms reporting neither are assigned ownership status based on pre-reform capital intensity, with those below (above) the median classified as tenants (owner-occupiers). The horizontal axis indicates the number of years since the reform. Each dot represents the point estimate of the treatment effect for that year, with the reference category set at -1. The vertical bands indicate the corresponding 95 percent confidence intervals. The regression specification includes fixed effects for sector, pre-reform size, and accounting month. The coefficients are estimated using a staggered Difference-in-Differences approach, following Callaway and Sant'Anna (2021). Standard errors are clustered at the firm level.

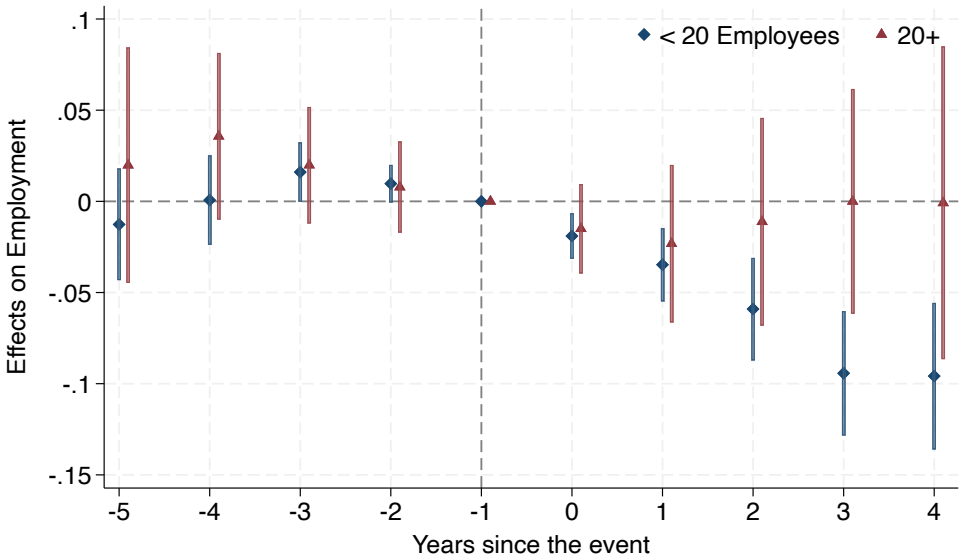

Figure A.20: Employment Effects: Heterogeneity by Size (Number of Formal Employees)

Notes: The figure shows the dynamic effects of the reform on formal employment, separately for different firm sizes based on their pre-reform formal employment. The outcome variable is number of workers, normalized by its pre-reform value, implying the coefficients can be interpreted as a percent changes. The horizontal axis indicates the number of years since the reform. Each dot represents the point estimate of the treatment effect for that year, with the reference category set at -1. The vertical bands indicate the corresponding 95 percent confidence intervals. The regression specification includes fixed effects for sector, pre-reform size, and accounting month. The coefficients are estimated using a staggered Difference-in-Differences approach, following Callaway and Sant'Anna (2021). Standard errors are clustered at the firm level.

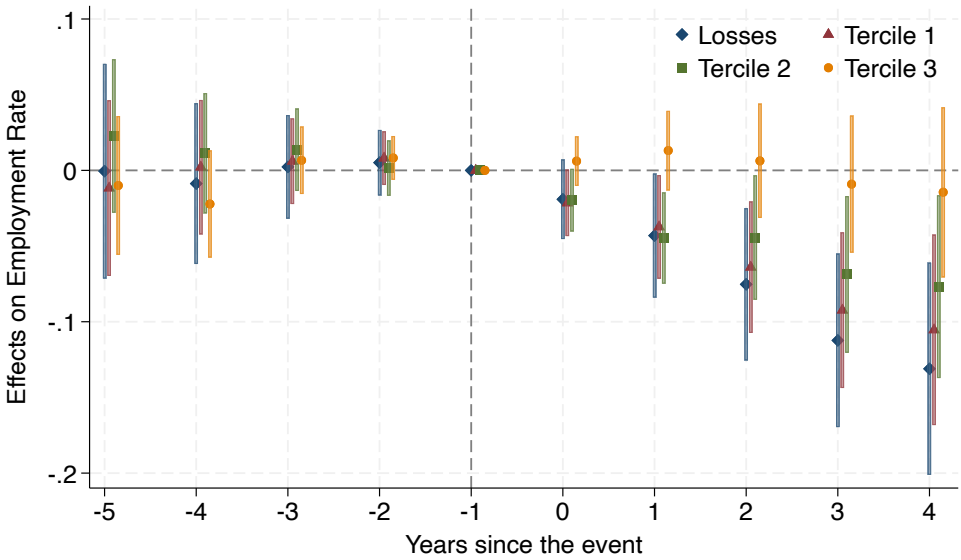

Figure A.21: Employment Effects: Heterogeneity by Profitability

Notes: The figure shows the dynamic effects of the reform on formal employment for firms at different points in the pre-reform profit margins (EBIT/revenues) distribution. The outcome variable is number of workers, normalized by its pre-reform value, implying the coefficients can be interpreted as a percent changes. The horizontal axis indicates the number of years since the reform. Each dot represents the point estimate of the treatment effect for that year, with the reference category set at -1. The vertical bands indicate the corresponding 95 percent confidence intervals. The regression specification includes fixed effects for sector, pre-reform size, and accounting month. The coefficients are estimated using a staggered Difference-in-Differences approach, following Callaway and Sant'Anna (2021). Standard errors are clustered at the firm level.

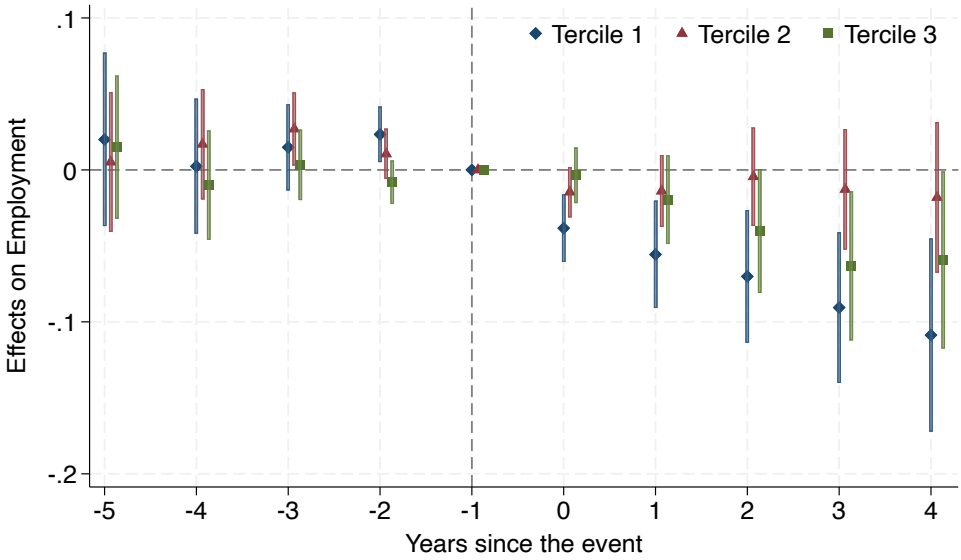

Figure A.22: Employment Effects: Heterogeneity by the Level of Leverage

Notes: The figure shows the dynamic effects of the reform on formal employment for firms with different levels of pre-reform leverage (debt-to-asset ratio). The outcome variable is number of workers, normalized by its pre-reform value, implying the coefficients can be interpreted as a percent changes. The horizontal axis indicates the number of years since the reform. Each dot represents the point estimate of the treatment effect for that year, with the reference category set at −1. The vertical bands indicate the corresponding 95 percent confidence intervals. The regression specification includes fixed effects for sector, pre-reform size, and accounting month. The coefficients are estimated using a staggered Difference-in-Differences approach, following Callaway and Sant'Anna (2021). Standard errors are clustered at the firm level.

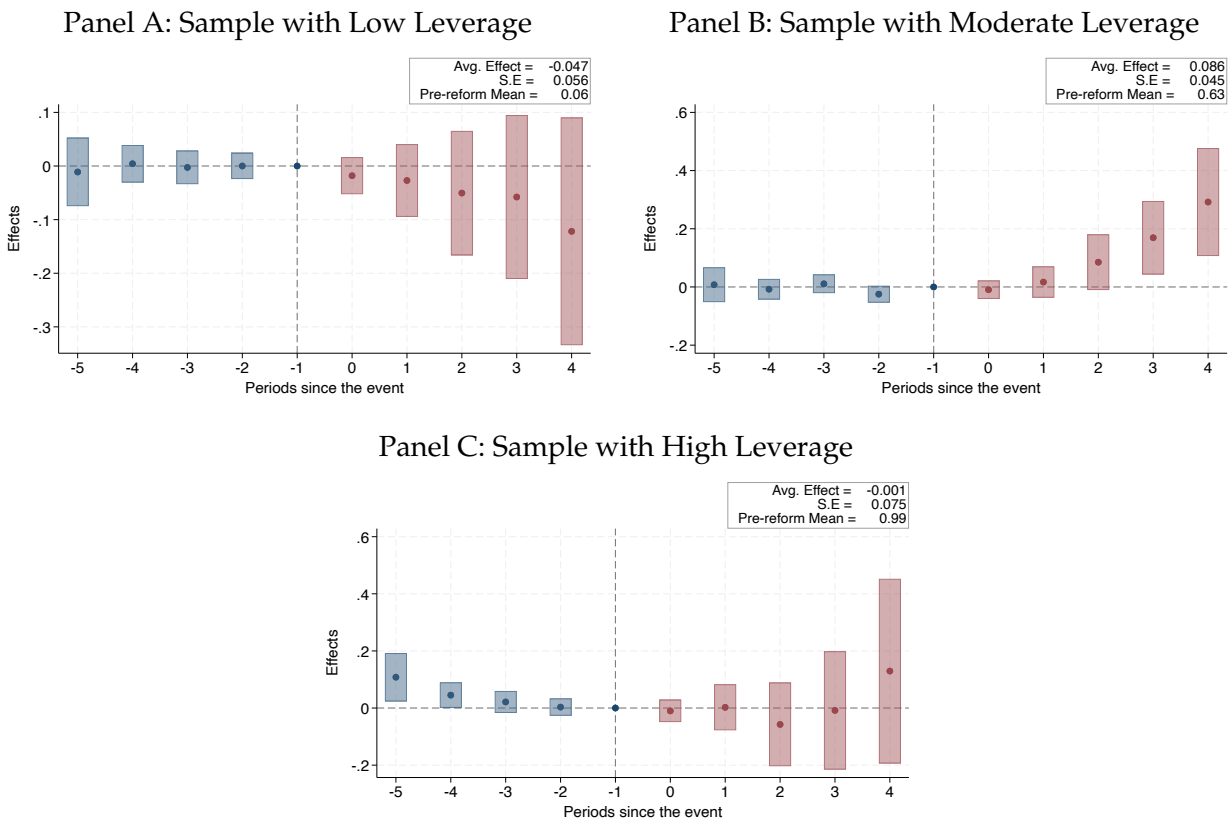

Figure A.23: Effects of the Reform on Firm's Debt

Notes: In all panels, the outcome variable is total debt scaled by pre-reform assets. In Panel A, I restrict the sample to firms in the first tercile of the pre-reform leverage distribution. Panel B restricts the sample to firms in the second tercile, while Panel C restricts the sample to firms in the third tercile (highly leverage firms). Each regression controls for the sector of operation, pre-reform size (by employment), accounting month, and occupancy status (Tenant or owner-occupier). The coefficients are estimated using a staggered Difference-in-Differences design following Callaway and Sant'Anna (2021), with not-yet treated units serving as the comparison group. Standard errors are clustered at the firm level

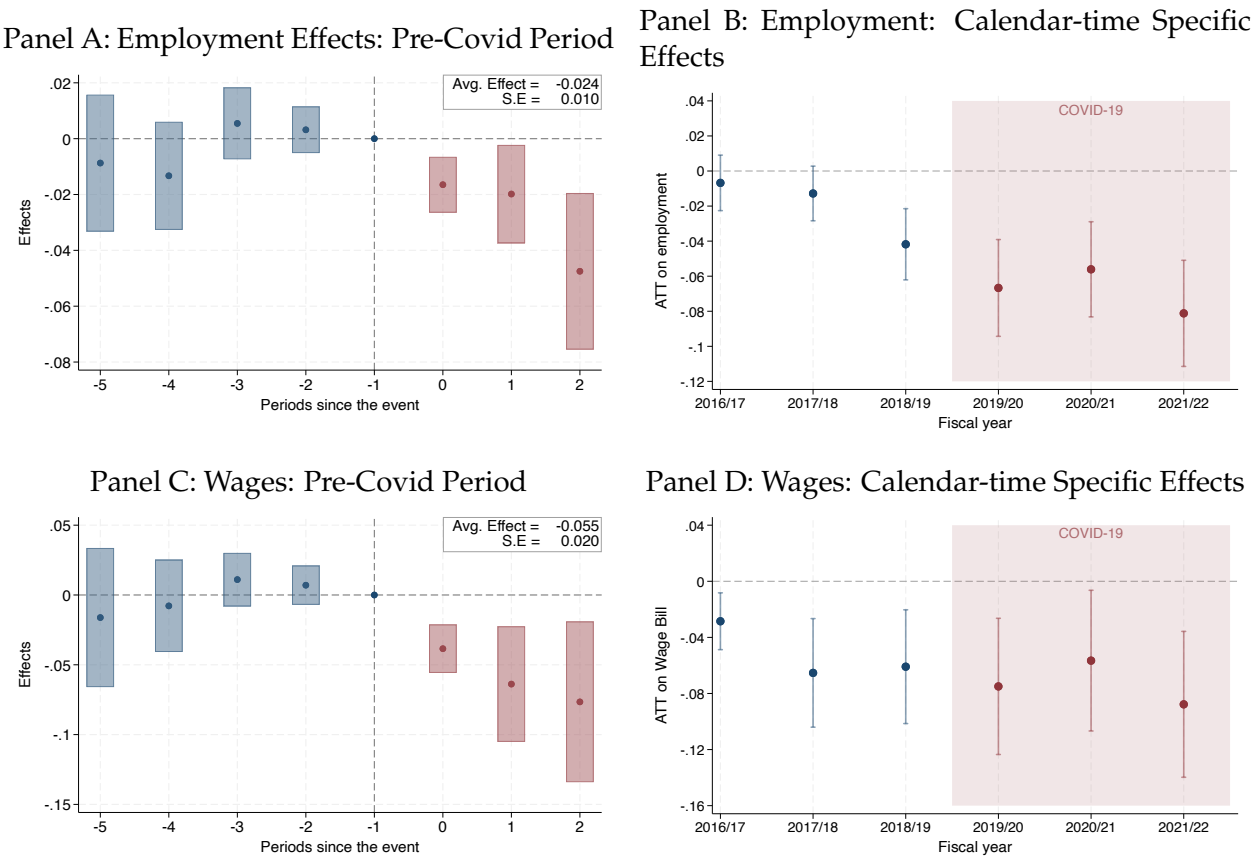

Figure A.24: Effects on Employment and Wages: Addressing COVID-19.

Notes: Panels A and C present the effects on wages and employment, while restricting the sample to pre-covid period, such that the last year in the sample is 2018/19. Panels B and D present the Calendar-time specific effects on employment and wages. The coefficients are estimated using a staggered Difference-in-Differences design following Callaway and Sant'Anna (2021), with not-yet treated units serving as the comparison group. Standard errors are clustered at the firm level

I.B. Additional Tables

Table A.1: Effects of the reform on Firms' Revenues, Profits, Capital, and Investment

	(1) Log Revenues	(2) Profits	(3) Log Capital	(4) Investment/Lagged Capital	(5) Investment: Extensive
ATT	0.013	-0.033*	-0.139**	-0.027**	-0.049**
	(0.042)	(0.019)	(0.065)	(0.011)	(0.022)
Sector of Operation	Y	Y	Y	Y	Y
Pre-reform Size	Y	Y	Y	Y	Y
Other Controls	Y	Y	Y	Y	Y
Observations	33012	33012	29097	43029	43029

Note: The table presents the average effects of the reform on firms' revenues, profits, capital, and investment. Profits are defined as earnings before interest and taxes (EBIT) (sales minus cost of sales, operating and administration expenses). These are scaled by pre-reform revenues. Investment is defined as the annual change in fixed assets, normalized by lagged fixed assets. The last column present extensive margin response of investment. The coefficients are estimated using a staggered Difference-in-Differences approach following Callaway and Sant'Anna (2021), with not-yet treated units used as the comparison group. Each regression includes fixed effects for sector, and pre-reform size. Under "Other Controls," I include dummies for accounting month and whether the firm filed returns on time, all measured in the pre-reform period. Standard errors are clustered at the firm level.

Table A.2: Effects on Wage Distribution: 25th, 25th, and 75th Percentiles

	(1) p25	(2) p50	(3) p75
ATE	-0.015	-0.064	-0.184**
	(0.025)	(0.045)	(0.082)
Sector of Operation	Yes	Yes	Yes
Pre-reform Size	Yes	Yes	Yes
Other Controls	Yes	Yes	Yes
Pre-reform Mean	0.372	0.496	0.816
Observations	19125	19125	19125

Note: The table presents the average effects of the reform on within-firm wage quantiles. The dependent variables are the 25th, 50th, and 75th within-firm percentiles of wages, scaled by the pre-reform formal wage bill, i.e., $\frac{q_{i,t}}{W_{i,0}} \times 100$, where $q_{i,t}$ denotes firm i's q^{th} wage quantile at time t and $W_{i,0}$ denotes firm i's pre-reform wage bill. Accordingly, the coefficient estimates should be interpreted as percentage-point changes relative to the pre-reform wage bill. The sample is restricted to firms with at least five workers in the pre-reform period. All models include fixed effects for sector of operation and pre-reform size. Under "Other Controls," I include dummies for accounting month, whether the firm reported owning land and buildings, and whether the firm filed returns on time, all measured in the pre-reform period. The coefficients are estimated using a staggered Difference-in-Differences approach following Callaway and Sant'Anna (2021). Standard errors are clustered at the firm level.

Table A.3: Effects on Wage Distribution: 20th, 40th, 60th, and 80th Percentiles

	(1) p20	(2) p40	(3) P60	(4) p80
ATE	-0.027	-0.034	-0.110*	-0.197**
	(0.026)	(0.033)	(0.064)	(0.099)
Sector of Operation	Yes	Yes	Yes	Yes
Pre-reform Size	Yes	Yes	Yes	Yes
Other Controls	Yes	Yes	Yes	Yes
Pre-reform Mean	0.351	0.439	0.583	0.950
Observations	19125	19125	19125	19125

Note: The table presents average effects of the reform on the wage distribution. The dependent variables are the 20th, 40th, 60th, and 80th within-firm percentiles of wages, scaled by the pre-reform formal wage bill, i.e., $\frac{q_{i,t}}{W_{i,0}} \times 100$, where $q_{i,t}$ denotes firm i's q^{th} wage quantile at time t and $W_{i,0}$ denotes firm i's pre-reform wage bill. Accordingly, the coefficient estimates should be interpreted as percentage-point changes relative to the pre-reform wage bill. The sample is restricted to firms with at least five workers in the pre-reform period. All models include fixed effects for sector of operation and pre-reform size. Under "Other Controls," I include dummies for accounting month, whether the firm reported owning land and buildings, and whether the firm filed returns on time, all measured in the pre-reform period. The coefficients are estimated using a staggered Difference-in-Differences approach following Callaway and Sant'Anna (2021). Standard errors are clustered at the firm level.

Table A.4: Property Taxes and Worker Composition

	(1) $< p25$	(2) p25-50	(3) p50-75	(4) p75-100
Average Effect (in pp)	0.404	0.698	2.212**	-2.838**
	(1.213)	(1.046)	(0.879)	(1.394)
Sector of Operation	Y	Y	Y	Y
Pre-reform Size	Y	Y	Y	Y
Other Controls	Y	Y	Y	Y
Pre-reform Mean	21.411	22.885	24.831	29.341
Observations	19125	19125	19125	19125

Note: The table presents the average effects of the reform on within firm employment composition. The dependent variables are employment shares in each quartile, relative to pre-reform total employment, i.e., $\frac{E_{i,b,t}}{E_{i,0}} \times 100$, where $E_{i,q,t}$ denotes firm i's number of workers in quartile q at time t and $E_{i,0}$ denotes firm i's total formal employment in the pre-reform period. Accordingly, the coefficient estimates should be interpreted as percentage-point changes relative to the pre-reform total employment. The sample is comprised of active firms with at least five workers in the pre-reform period. All models include fixed effects for sector of operation and pre-reform size. Under "Other Controls," I include dummies for accounting month, whether the firm reported owning land and buildings, and whether the firm filed returns on time, all measured in the pre-reform period. The coefficients are estimated using a staggered Difference-in-Differences approach following Callaway and Sant'Anna (2021). Standard errors are clustered at the firm level.

Table A.5: Implications for Central Government Tax Revenue

	CIT		Employee Tax		Combined	
	(1)	(2)	(3)	(4)	(5)	(6)
ATE	-0.085	-0.091	-0.543***	-0.451***	-0.627***	-0.542**
	(0.109)	(0.129)	(0.126)	(0.141)	(0.188)	(0.217)
Controls	No	Yes	No	Yes	No	Yes
Pre-reform Mean	1.455	1.455	6.355	6.355	7.810	7.810
Observations	137862	137862	137862	137862	137862	137862
Jobs Lost per UGX 1					0.192	0.166

Note: The table presents the average effects of the reform on central government tax revenue. All the analysis is aggregated at firm level. The outcome variables are in millions of Uganda shillings (UGX), constant 2016/17. Columns 1 and 2 show the effects on corporate income tax, columns 3 and 4 show the effects on employee tax revenue (PAYE), while columns 5 and 6 show the effects on the combined revenue. The sample is a panel of all firms that were active in the pre-reform year, replacing zero for the tax revenue if they exit. The controls include sector of operation, accounting month, and a dummy for occupancy status (tenant or owner-occupier). Standard errors are clustered at the firm level.

II. APPENDIX: MODELS

II.A. Theoretical Framework: Detailed Steps

Landlord's Maximization Problem:— The landlord sets rent to maximize his expected utility:

$$\max_R \; q(R)\,(R-T) + (1-q(R))\,(-p_e T)\,, \tag{12}$$

where $q(R)$ denotes the probability that the unit is filled at rent R, $p_e \leq 1$ is the probability that the tax is enforced given that unit is vacant. The first term the landlord's problem is the expected payoff if the unit is filled: the landlord earns $R-T$ with probability $q(R)$. The second term is the expected payoff if the unit remains vacant: the landlord pays $p_e T$ in expected tax with probability $1-q(R)$.

Expanding (12):

$$\max_R \; q(R)R - q(R)T - p_e T + q(R)p_e T \tag{13}$$

$$= \max_R \; q(R)R - q(R)(1-p_e)T - p_e T \tag{14}$$

The last term, $-p_e T$, does not depend on R and drops out of the optimization. The landlord effectively maximizes:

$$\max_R \; q(R)\,[R-(1-p_e)T] \tag{15}$$

The landlord behaves as if facing a per-unit cost of $(1-p_e)T$ for each occupied unit. This cost is the additional expected tax the landlord incurs by having the unit occupied rather than vacant. Under universal enforcement ($p_e = 1$), this cost is zero— the landlord pays T regardless of occupancy, so occupancy triggers no additional liability. Under fully selective enforcement ($p_e = 0$), the cost is the full tax T— each occupied unit triggers the entire tax bill.

The first-order condition of (15) is:

$$q'(R)\,[R-(1-p_e)T] + q(R) = 0 \tag{16}$$

Solving for R:

$$\boxed{R^* = (1-p_e)T - \frac{q(R^*)}{q'(R^*)}} \tag{17}$$

The optimal rent has two components. The first, $(1-p_e)T$, is the effective marginal cost of occupancy—the portion of the tax that the landlord can avoid by leaving the unit vacant. The second, $-q(R^*)/q'(R^*)$, is the standard markup that depends on the shape

of the demand curve. Under universal enforcement ($p_e = 1$), the first term vanishes and the optimal rent depends only on demand conditions, not on the tax, recovering the neoclassical result.

Characterizing the Pass-through Rate:— Defining $c \equiv (1 - p_e)T$ as the effective marginal cost of occupancy, the landlord's first-order condition implicitly defines the optimal rent R^* as a function of the effective cost c, and hence of T. Define the function:

$$G(R, c) \equiv q'(R)(R - c) + q(R) = 0 \tag{18}$$

By the implicit function theorem:

$$\frac{dR^*}{dc} = -\frac{\partial G/\partial c}{\partial G/\partial R} \tag{19}$$

Computing the partial derivatives:

$$\frac{\partial G}{\partial c} = -q'(R)$$

$$\frac{\partial G}{\partial R} = q''(R)(R - c) + q'(R) + q'(R) = q''(R)(R - c) + 2q'(R)$$

Therefore:

$$\frac{dR^*}{dc} = -\frac{-q'(R^*)}{q''(R^*)(R^* - c) + 2q'(R^*)} = \frac{q'(R^*)}{q''(R^*)(R^* - c) + 2q'(R^*)} \tag{20}$$

From the first-order condition (16), we can substitute $R^* - c = -q(R^*)/q'(R^*)$:

$$\frac{dR^*}{dc} = \frac{q'}{q''\left(-\frac{q}{q'}\right) + 2q'} = \frac{q'}{-\frac{q\,q''}{q'} + 2q'} = \frac{(q')^2}{-(q)(q'') + 2(q')^2} \tag{21}$$

Dividing numerator and denominator by $(q')^2$:

$$\frac{dR^*}{dc} = \frac{1}{2 - \frac{q(R^*)\,q''(R^*)}{[q'(R^*)]^2}} \tag{22}$$

Since $c = (1 - p_e)T$ and hence $dc/dT = (1 - p_e)$, the pass-through rate with respect to the tax liability is:

$$\boxed{\rho = \frac{dR^*}{dT} = \frac{dR^*}{dc} \cdot \frac{dc}{dT} = \frac{1 - p_e}{2 - \theta}}, \tag{23}$$

where $\theta \equiv \frac{q(R^*)\, q''(R^*)}{[q'(R^*)]^2}$ is the demand curvature parameter introduced in the tax incidence literature by Weyl and Fabinger (2013).

In the pass-through formula (23), we require $\theta < 2$ for $\rho \geq 0$. This restriction follows from the second-order condition of the landlord's maximization problem, which ensures that the optimal rent R^* is a well-defined interior maximum. To see this, note that the second-order condition requires:

$$\frac{d^2}{dR^2}\left[q(R)(R - c)\right] = q''(R)(R - c) + 2q'(R) < 0 \tag{24}$$

From the first-order condition (16), we can substitute $R^* - c = -q(R^*)/q'(R^*)$:

$$q''(R^*)\left(-\frac{q(R^*)}{q'(R^*)}\right) + 2q'(R^*) = -\frac{q(R^*)\, q''(R^*)}{q'(R^*)} + 2q'(R^*) \tag{25}$$

Factoring out $q'(R^*)$:

$$= q'(R^*)\left(2 - \frac{q(R^*)\, q''(R^*)}{[q'(R^*)]^2}\right) = q'(R^*)\,(2 - \theta) \tag{26}$$

Since $q'(R^*) < 0$, the expression in (26) is negative if and only if $2 - \theta > 0$, i.e., $\theta < 2$. When $\theta \geq 2$, the landlord's profit function is convex at the candidate solution, and no interior optimum exists.

III. APPENDIX: BACK-OF-THE-ENVELOPE CALCULATIONS

III.A. Decomposing the PAYE Decline: Informalization vs. Job Destruction

To assess how much of the decline in central government PAYE revenues is attributable to wage informalization rather than other margins, I First, estimate the implied PAYE loss from informalization alone, and the express this implied loss as a share of the total estimated PAYE decline to recover the fraction accounted for by the informalization margin. The exercise combines three objects directly observable in the data — the informal wage share, the pre-reform CIT wage bill, and the average effective PAYE rate — with two reduced-form estimates from the main analysis: the average treatment effect on wage informality and the average treatment effect on PAYE revenues.

The *informal wage share* of firm i at time t is defined as the fraction of the CIT-reported wage bill that does not appear on employee-level returns:

$$s_{it}^{inf} = \frac{W_{it}^{CIT} - W_{it}^{PAYE}}{W_{it}^{CIT}} \tag{27}$$

where W_{it}^{CIT} denotes the total wage bill reported on the firm's corporate income tax return, and W_{it}^{PAYE} denotes the total wage bill derived from employee-level Pay-As-You-Earn (PAYE) returns for the same firm and year. The difference between the two captures wages paid to workers who are not registered in the formal payroll system. As documented in Section VI.C, the reform increased s_{it}^{inf} by **3.3** percentage points on average among treated firms.

The *pre-reform CIT wage bill*, $W_{i,pre}^{CIT}$, is defined as firm i's average CIT-reported wage bill in the years prior to the reform. This serves as the base to which the informalization increase is applied, converting the estimated percentage-point increase in the informal wage share into shillings worth of wages shifting out of the formal payroll.

The *average effective PAYE rate*, τ^{PAYE}, captures the fraction of the formal wage bill that is collected as PAYE tax. It is computed at the firm level as:

$$\tau_i^{PAYE} = \frac{\text{PAYE}_{i,pre}}{W_{i,pre}^{PAYE}} \tag{28}$$

where $\text{PAYE}_{i,pre}$ is the firm's average PAYE payment in the pre-reform period and $W_{i,pre}^{PAYE}$ is the corresponding average formal wage bill from employee-level returns. This rate converts the loss in the formal wage base into an implied loss in PAYE tax revenues.

Combining these three objects, the implied PAYE revenue loss attributable to infor-

malization alone for firm i is:

$$\text{Implied PAYE loss}_i = \hat{\beta}^{inf} \times W_{i,pre}^{CIT} \times \tau_i^{PAYE} \tag{29}$$

where $\hat{\beta}^{inf}$ is the reduced form estimate on wage informality presented in Section VI.C. The intuition is straightforward: $\hat{\beta}^{inf} \times W_{i,pre}^{CIT}$ gives the shillings worth of wages that shifted from formal to informal arrangements as a result of the reform, and multiplying by τ_i^{PAYE} converts this wage base loss into the implied tax revenue loss.

Denoting the average treatment effect on PAYE tax revenues as $\hat{\delta}^{PAYE}$, we can define the share of PAYE decline accounted for by informalization as:

$$\text{Share explained} = \frac{\frac{1}{N_T} \sum_{i \in \mathcal{T}} \text{Implied PAYE loss}_i}{\hat{\delta}^{PAYE}} \tag{30}$$

where $\mathcal{T}$ denotes the set of treated firms and N_T is the number of treated firms. $\hat{\delta}^{PAYE}$ is the reduced form estimate on PAYE in Section VIII. As this ratio gets close to one, wage informalization within treated firms fully accounts for the PAYE tax revenue decline.